

FPT University HCM



Deliverable 4

Topic: Agriculture UAV

**A platform for managing and
coordinating regional precision
agricultural spraying drone
services.CopperMekong River**

SYSTEM DESIGN SPECIFICATION (SDS)

Subject: ITA301

Instructor: *Ngo Dang Ha An*

Ho Chi Minh City, July 2026

PROJECT INFORMATION

1. **Department responsible:** This project was carried out by a group of students currently studying at FPT University Ho Chi Minh City as part of their ITA301 – Systems Analysis and Design course.

List of Project Members

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2. **Content to be implemented:** Research and analysis of a platform supporting the coordination of precision agricultural spraying drones aims to optimize UAV management, spray scheduling, task progress monitoring, and support the optimization of operational models and management of precision agricultural services.

Research project objectives:

- Optimize human resources, minimize scheduling conflicts and coordination errors.
 - Increase the efficiency of pesticide spraying operations using UAVs.
 - Monitoring task progress at a visual, operational level and enhancing the transparency of log data.
 - Contributing to the advancement of digital transformation in the field of precision agriculture in Vietnam.
3. **Execution time:** May 2026 – July 2026

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1. Introduction

1.1 Purpose: Define the objectives of the system design document.

This system design document aims to provide a detailed description of the system's architecture, data structures, and functional behavior. Agriculture UAV.

The documentation serves as a primary reference point for the development team during system deployment, and also assists the testing team in building appropriate test cases to ensure the system fully meets the defined business requirements.

1.2 Scope: System scope Agriculture UAV

1.2.1 In-scope:

- User account management and role-based access control.
- Managing land plot records and user service requests.
- Managing and developing electronic prescriptions for pesticide spraying operations.
- Mission coordination, including personnel assignment and UAV resource management.
- Monitor task progress throughout the operational process.
- Recording operational logs includes: OperationLog and Spray Log
- Manage maintenance operations for UAV equipment, batteries, and charging stations.
- Managing invoices and compiling reports to support business operations.

1.2.2 Out-of-scope:

- Direct flight control or automated UAV flight control.

- AI systems for pest and disease identification or agricultural image analysis.
- Processing real-time telemetry data from UAVs.
- Managing UAV firmware or hardware systems.

1.3. Modeling Tools: List the tools used

- **Lucidchart:** Use it to create data flow diagrams (DFDs), use case diagrams, and activity diagrams thanks to its excellent online collaboration features.
- **Enterprise Architect:** Used for designing class diagrams and sequence diagrams to ensure compliance with the UML 2.0 standard.
- **Draw.io (diagrams.net):** Use this to quickly sketch out logical structure diagrams during group discussions.

1.4. Diagram List: A numbered list of diagrams

1. **Architecture Diagram:** 3-Layer (Presentation – Application – Data Layer) architecture diagram of the Agriculture UAV system.
2. **UI Wireframes:** A user interface design sketch for the system's main screens, tailored to each corresponding role (Actor).
3. **Physical Data Model:** The physical database design diagram includes 15 tables with PK/FK, data types, and referential integrity constraints.
4. **Threat & Mitigation Matrix:** A table analyzing security threats and corresponding remedial measures.
5. **Design Traceability Matrix:** The traceability table connects Functional Requirements to UI, Database Tables, and Test Cases.

2. Design Goals and Principles

System **Agriculture UAV – Precision Spraying Management Platform** It is designed with objectives to ensure usability, security, scalability, and maintainability during operation. The following design principles are applied:

2.1 Usability

The system aims to provide an intuitive interface and easy to use it caters to various user groups such as Administrator, Manager, Farmer, Agronomist, Drone Operator, and Technician. Functions are organized by role to reduce the number of steps and help users complete tasks quickly. The system also provides clear notifications, data entry instructions, and error messages to minimize errors during use.

2.2 Security

Security is a critical requirement of the system because the data includes user information, flight missions, and operational logs. The system employs user authentication through secure login, role-based access control (RBAC), password encryption using BCrypt, and JWT for

session management. Additionally, the system validates input data to mitigate risks such as SQL Injection and Cross-Site Scripting (XSS), and records audit logs for critical operations.

2.3 Maintainability

The system is designed with a 3-layer architecture (Presentation Layer, Application Layer, and Data Layer), separating the interface, business logic, and data. This organization facilitates maintenance, bug fixing, and the addition of new functionality without impacting the entire system. The source code is organized into modules to increase reusability and convenience during development.

2.4 Traceability

All functional requirements are linked to use cases, user interface (UI), database design, and test cases through the Design Traceability Matrix. This ensures that each requirement is implemented in the design and fully tested, while also supporting tracking of changes or bug discoveries.

2.5 Scalability

The system is built with future scalability in mind. Its modular architecture allows for the addition of features such as AI pest detection, IoT device integration, real-time telemetry data, or expansion to multiple farming areas without significantly altering the existing structure. The relational database is also standardized to support data expansion and increased user capacity.

2.6 Auditability

To ensure transparency and traceability, the system maintains comprehensive audit logs for critical activities such as logins, data changes, task assignments, and system administration. These logs are accessible only to administrators and support the testing, troubleshooting, and management and audit processes.

3. User Personas and User Journey

3.1 User Persona

Person:Farmer

Attribute	Description
Name	Nguyen Van Minh
Age	45
Occupation	Rice Farmer

Location	Dong Thap, Mekong Delta, Vietnam
Experience	Over 20 years in rice cultivation
Digital Skills	Basic smartphone and web application usage
Goals	Quickly request UAV spraying services, monitor mission progress, receive accurate spraying reports, and reduce manual farming effort.
Pain Points	Difficult to schedule spraying services during peak seasons, limited visibility of request status, and lack of real-time mission updates.
Needs	An easy-to-use platform that allows online service requests, mission tracking, notifications, and access to spraying reports.

3.2 User Journey

Primary Actor:Farmer

Related Use Case: UC-05 – Create Service Request

Step	User Action	System Response	Expected Outcome
1	Log into the system	Authenticate user credentials and display the dashboard	Farmer successfully accesses the system
2	Select Create Service Request	Display the service request form	Ready to submit a new request
3	Enter field information, crop details, pest conditions, and preferred spraying date	Validate required fields and input format	Valid information is accepted
4	Submit the request	Save request to the database and assign Pending status	Service request is successfully created
5	Wait for Agronomist review	Notify Agronomist for prescription review	Request enters the review process
6	Track request status	Display current request and mission status	Farmer can monitor request progress
7	Receive completion	Display spraying result	Farmer reviews the completed

	notification and report	and mission report	service information
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4. UI Wireframes

This section presents the main interface sketches of the Agriculture UAV Management System. Each interface illustrates how actors interact with the system to perform corresponding tasks. The screens are designed to ensure traceability between the Use Case, Functional Requirement, and interface design.

4.1 UI-01 Account Management

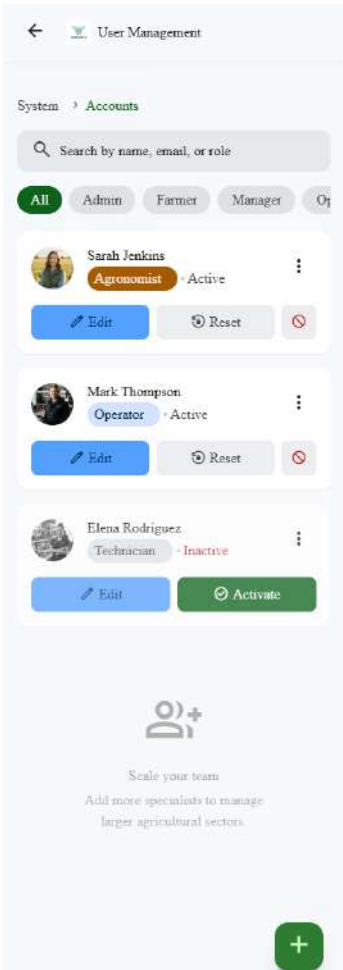
Related Use Cases:UC-01 – Centralized Account Management

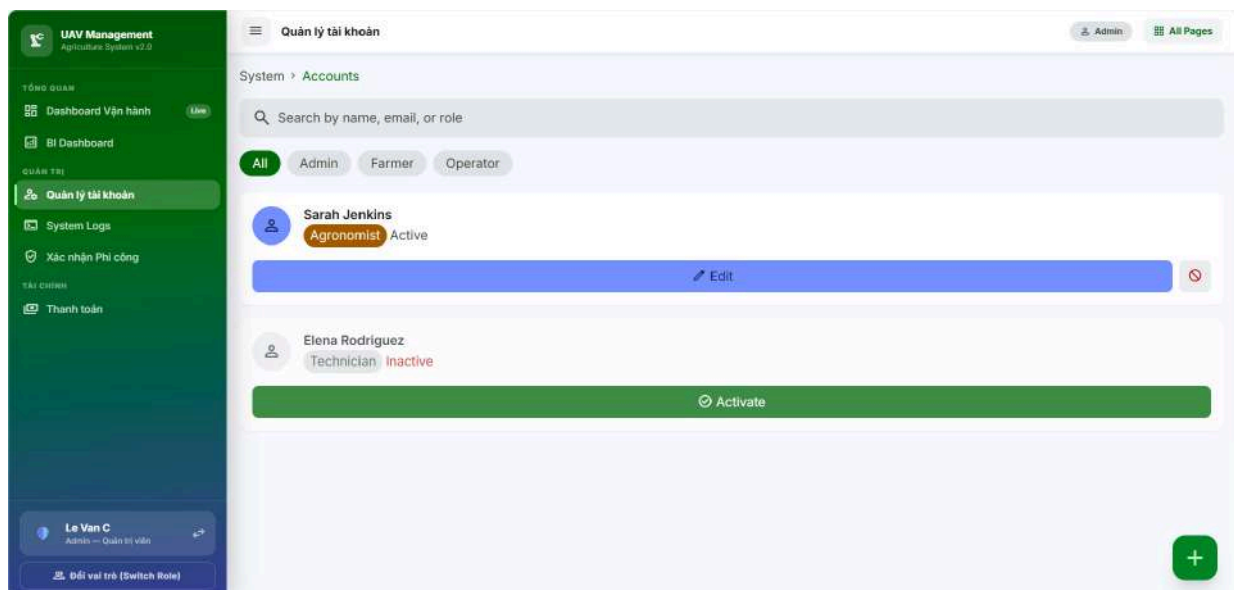
Actor: Administrator

Describe: Administrators manage user accounts, assign permissions by role, enable or disable accounts, and reset passwords when necessary.

Main ingredients

- Account list
- Search for users
- Role filter
- Account information
- Add account
- Edit account
- Delegation of authority
- Enable/Disable
- Reset password





4.2 UI-02 System Log

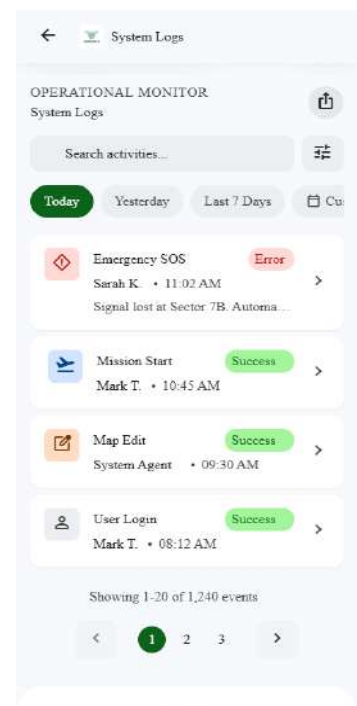
Related Use Cases: UC-02 – Operation monitoring and fault lookup

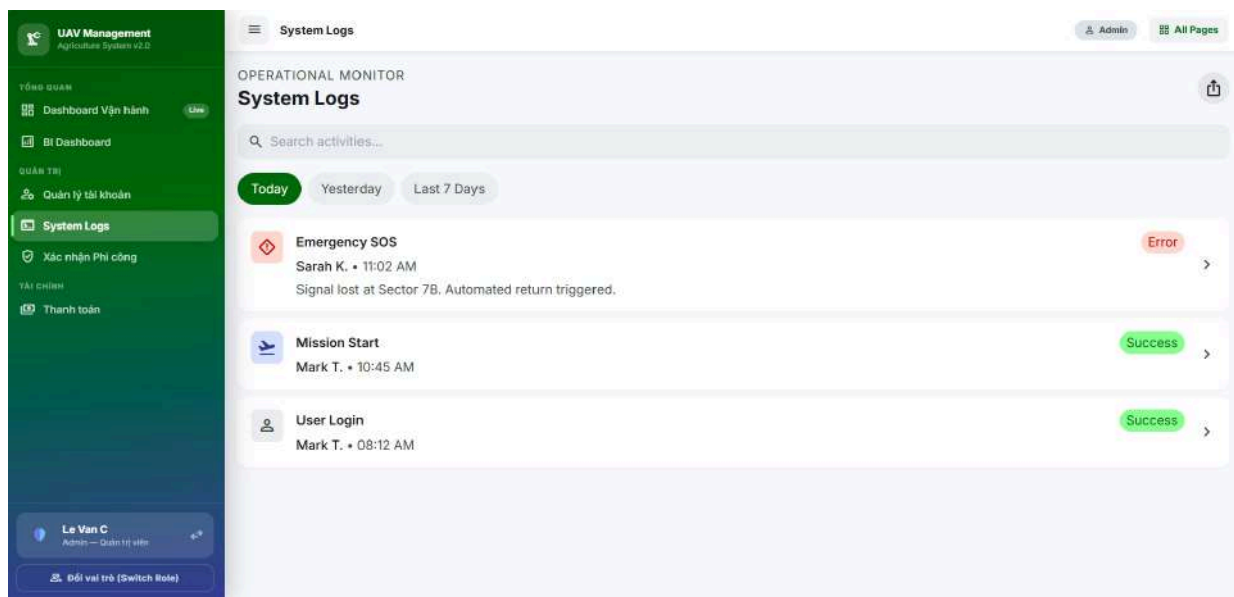
Actor: Administrator

Describe: Administrators monitor system activities, review logs, and check for errors to support administration and ensure system security.

Main ingredients

- System log
- User filter
- Active type filter
- Time filter
- Error details
- Search
- Export log





4.3 UI-03 Login Screen

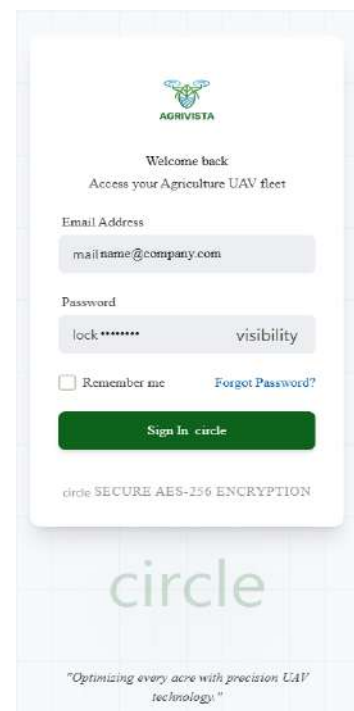
Related Use Cases: UC-03 – Login and security authentication

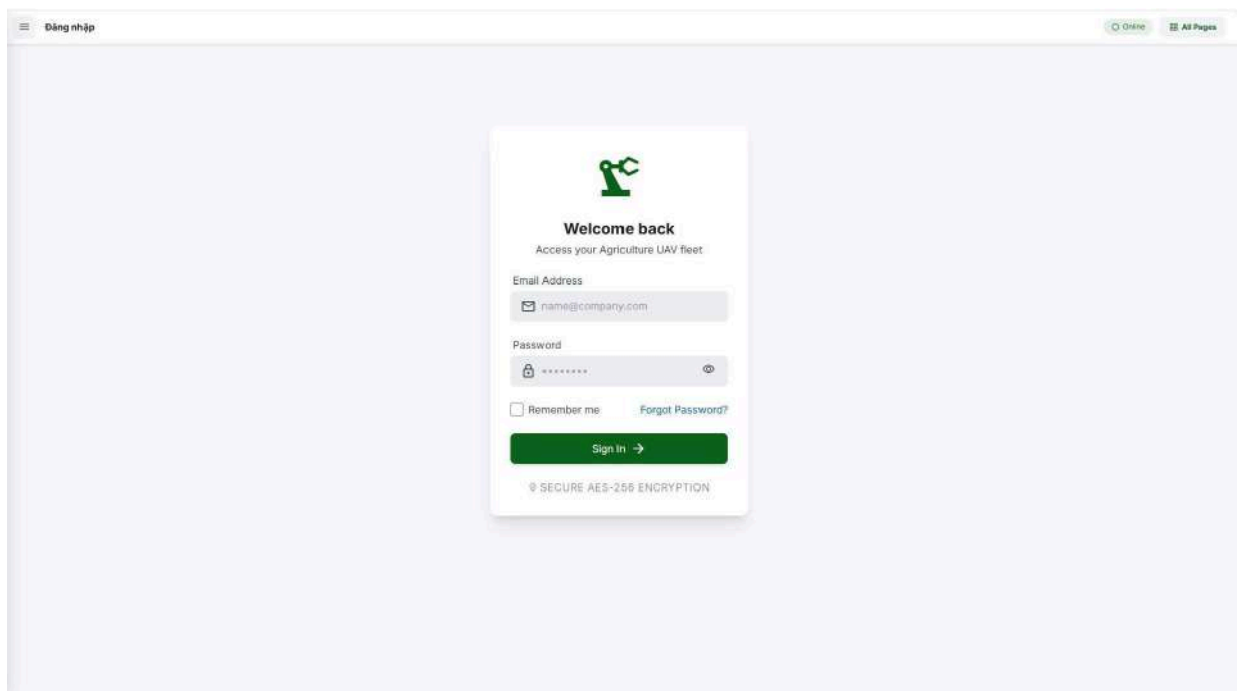
Actor: All users

Describe: Users log into the system using their email and password. After successful authentication, the system redirects them to an interface appropriate to their role.

Main ingredients

- System logo
- Email
- Password
- Remember me
- Login button
- Forgot password
- Error message





4.4 UI-04 Land Map and Record Management

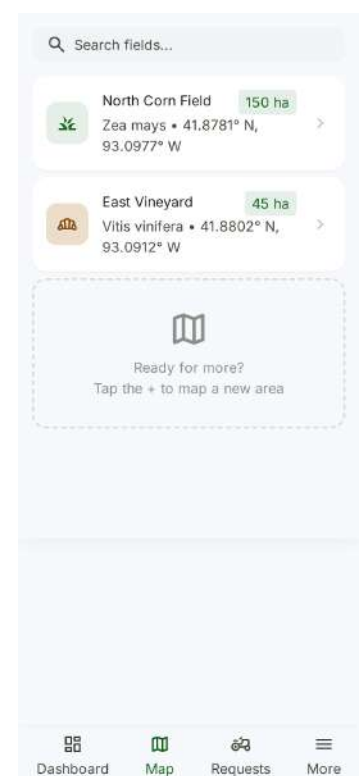
Related Use Cases: UC-04 – Land Map and Record Management

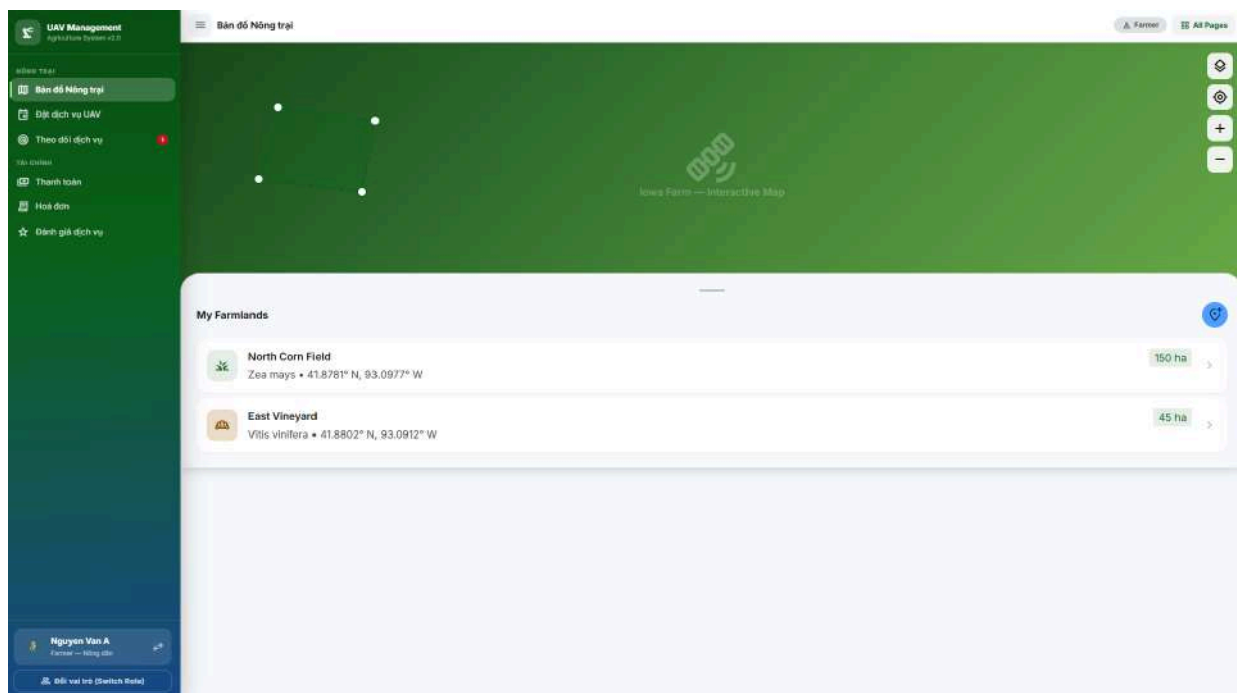
Actor: Farmer

Describe: The farmer manages information about the land plot, including its location, area, crop type, and cultivation history.

Main ingredients

- Map
- List of land plots
- Land plot information
- Type of crop
- GPS coordinates
- Edit
- Save





4.5 UI-05 Schedule Service

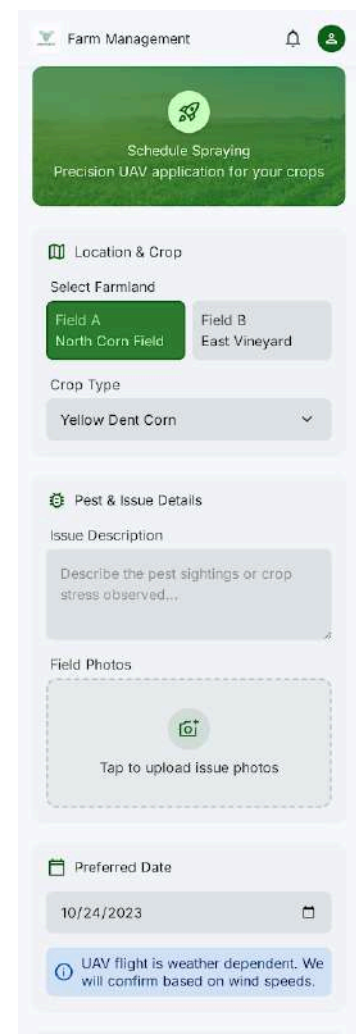
Related Use Cases: UC-05 – Schedule pest control service

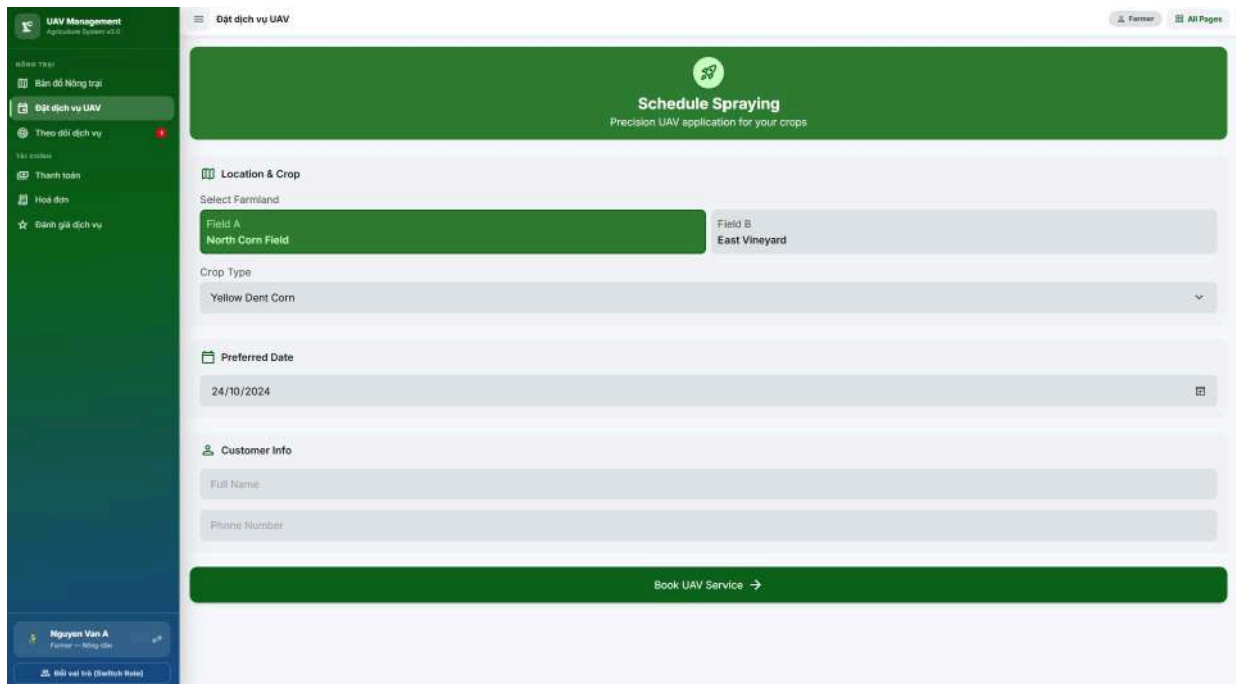
Actor: Farmer

Describe: Farmers request spraying services by entering information about their farming area, pest and disease status, and desired time frame.

Main ingredients

- Customer information
- Choose a plot of land
- Type of crop
- Description of pests and diseases
- Desired date
- Download image
- Submit a request
- Cancel





4.6 UI-06 Task Coordination

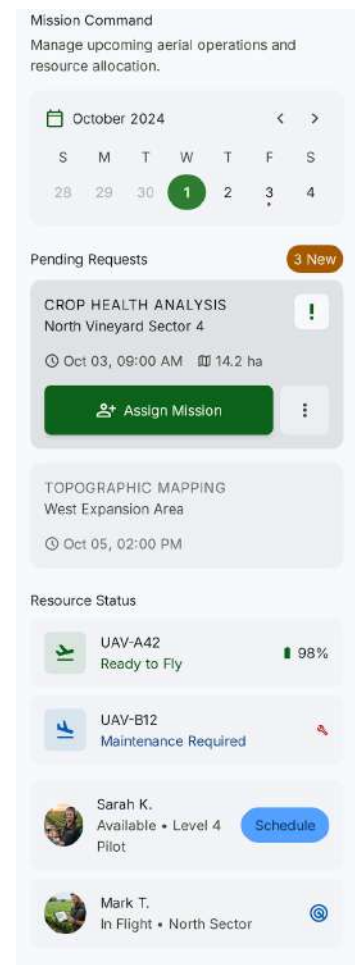
Related Use Cases: UC-06 – Flight schedule coordination and resource monitoring

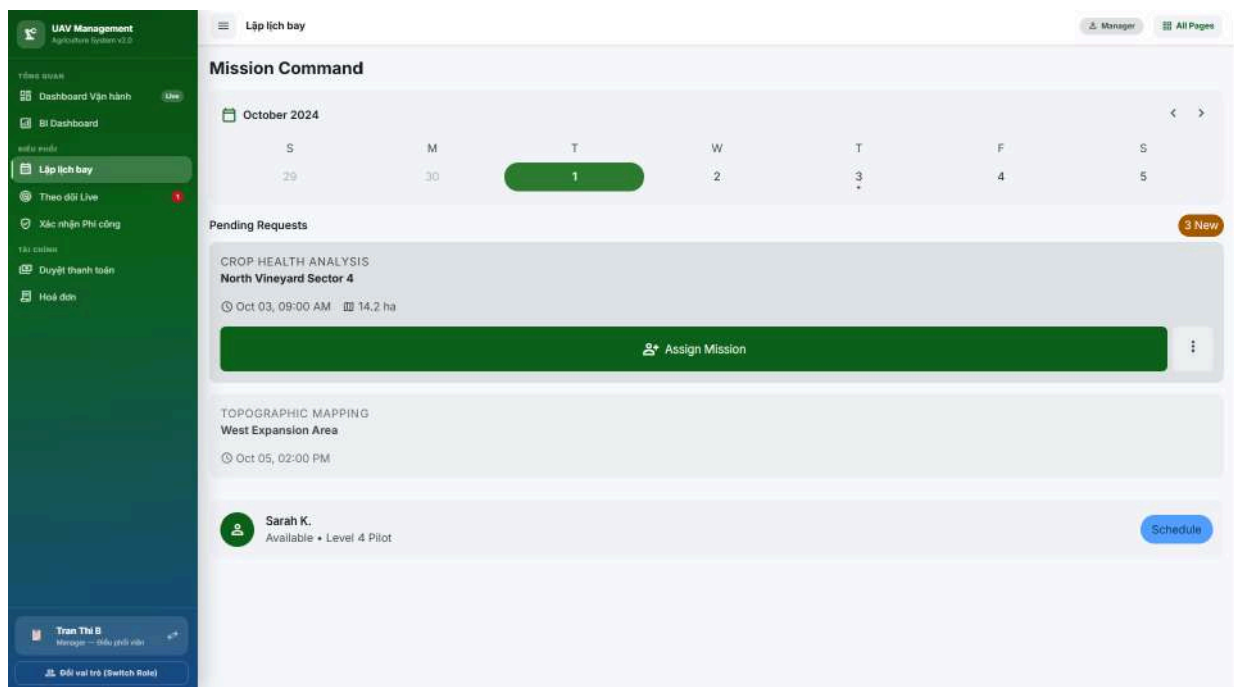
Actor: Manager

Describe: The manager assigns drones, pilots, and mission schedules, and detects scheduling conflicts.

Main ingredients

- List of pending tasks
- Drones available
- Available pilots
- Implementation schedule
- Conflict alert
- Assignment Button





4.7 UI-07 Dashboard Monitoring

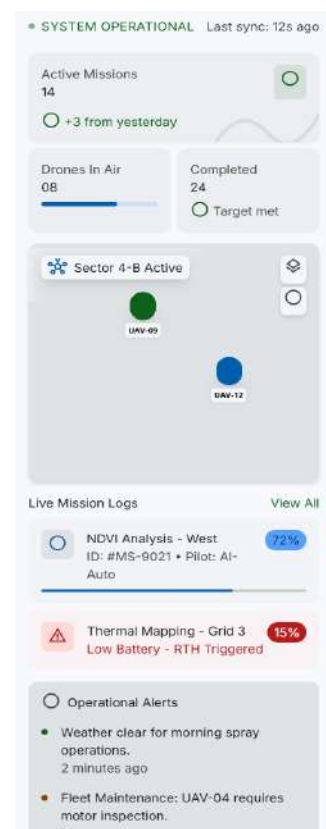
Related Use Cases: UC-06 – Flight schedule coordination and resource monitoring

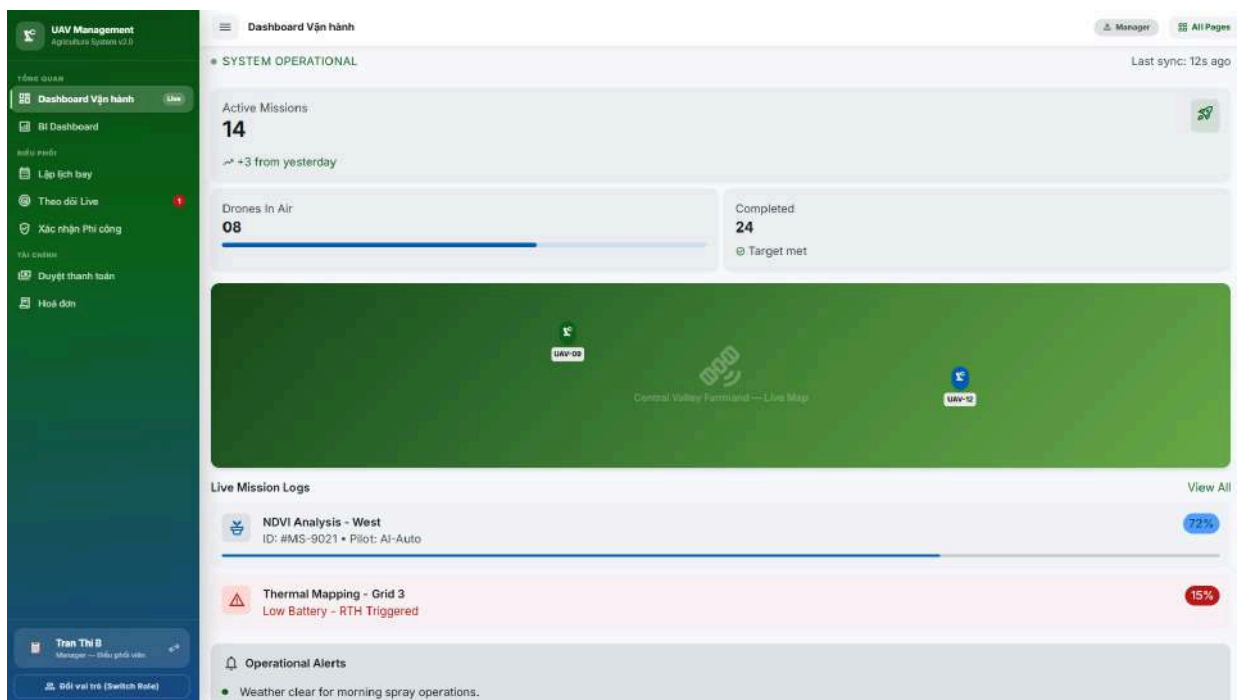
Actor: Manager

Describe: The dashboard displays the drone's operational status, mission progress, and resource usage in real time.

Main ingredients

- KPI Card
- Drone status
- Task progress
- Resource status
- Notification
- Monitoring map





4.8 UI-08 Electronic Prescription Device

Related Use Cases: UC-07 – Drug Request and Prescription Review

Actor: Agronomist

Describe: The agronomist reviews the service request and creates a suitable electronic prescription.

Main ingredients

- Information required
- Crop condition
- Recommended medication
- Dosage
- Approve
- Refuse

SERVICE REQUEST URGENT REVIEW

Maize Blight Mitigation
Field Sector 4G • West Valley Plantation •
requested by Operator Marcus Thorne

Crop Condition Detected: 14% Loss Risk

STRESS LEVEL
High (Level 4)

SOIL MOISTURE
42% (Normal)

i Multispectral imagery from DJI Mavic 3M indicates rapid spread towards the eastern boundary. Immediate application recommended.

Target Area

Sector 4G Boundary
12.4 Hectares

Digital Prescription

Recommended Fungicide
Quilt Xcel™ Fungicide

Application Rate (L/ha) 2.5

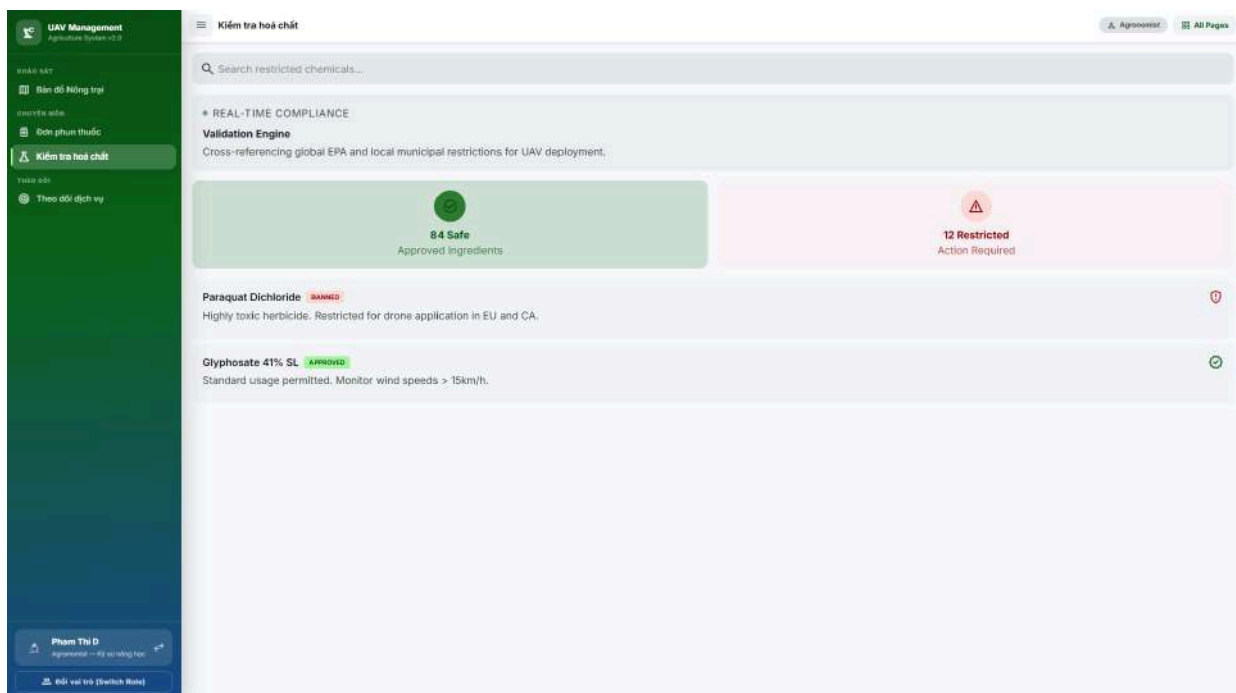
0.5 L

 Max 5.0 L

Flight Protocol Notes

Add specific spray altitude or wind speed constraints...

APPROVE & SYNC TO UAV
REJECT REQUEST



4.10 UI-10 Device Catalog Management

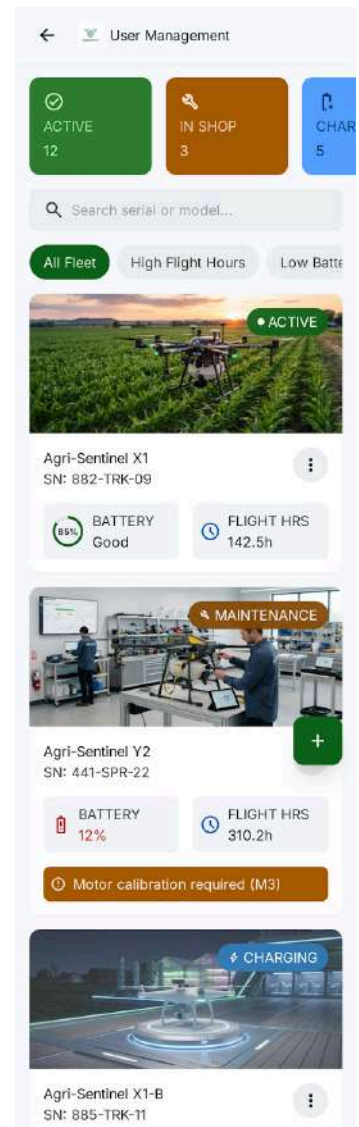
Related Use Cases: UC-09 – Drone and Battery Catalog Management

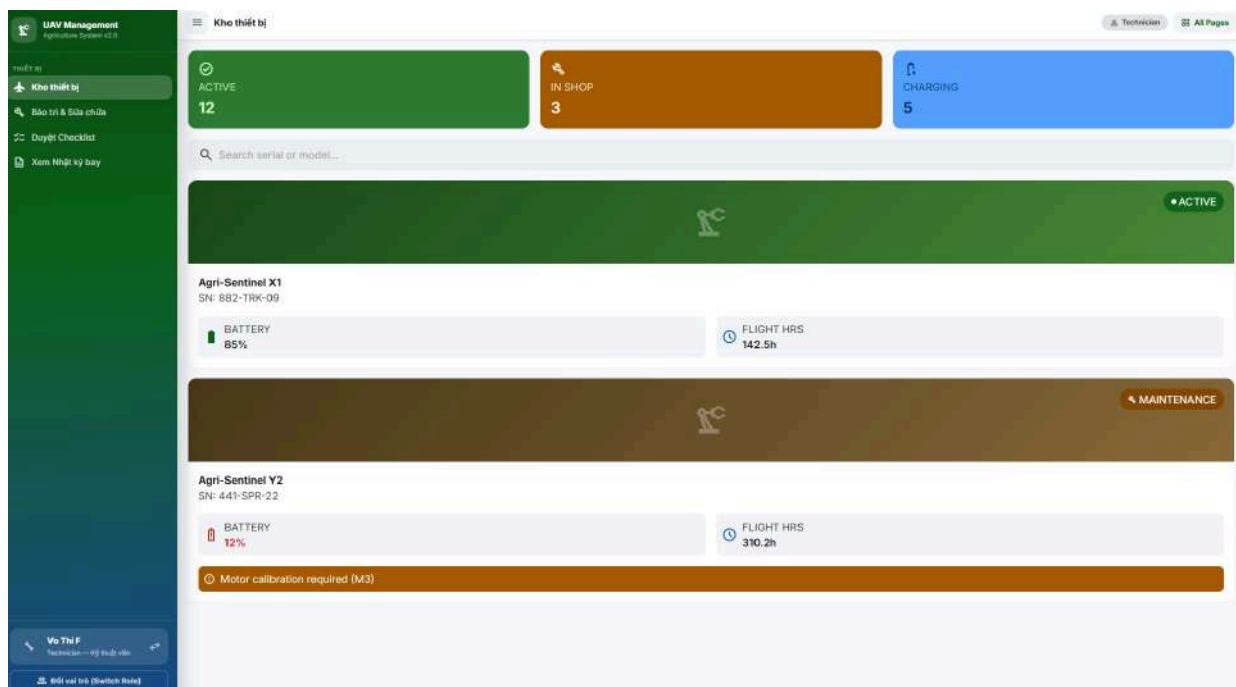
Actor: Technician

Describe: Technicians manage the drone and battery inventory, update equipment information, monitor operational status, and ensure equipment is always ready for flight missions.

Main ingredients

- Drone List
- Battery status
- Flight times
- Search
- Filter
- Update status





4.11 UI-11 Maintenance Management

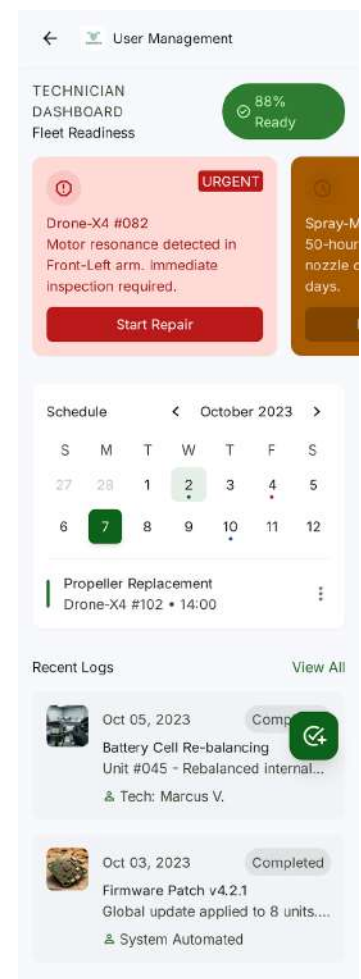
Related Use Cases: UC-10 – Automated Maintenance Management and Alert System

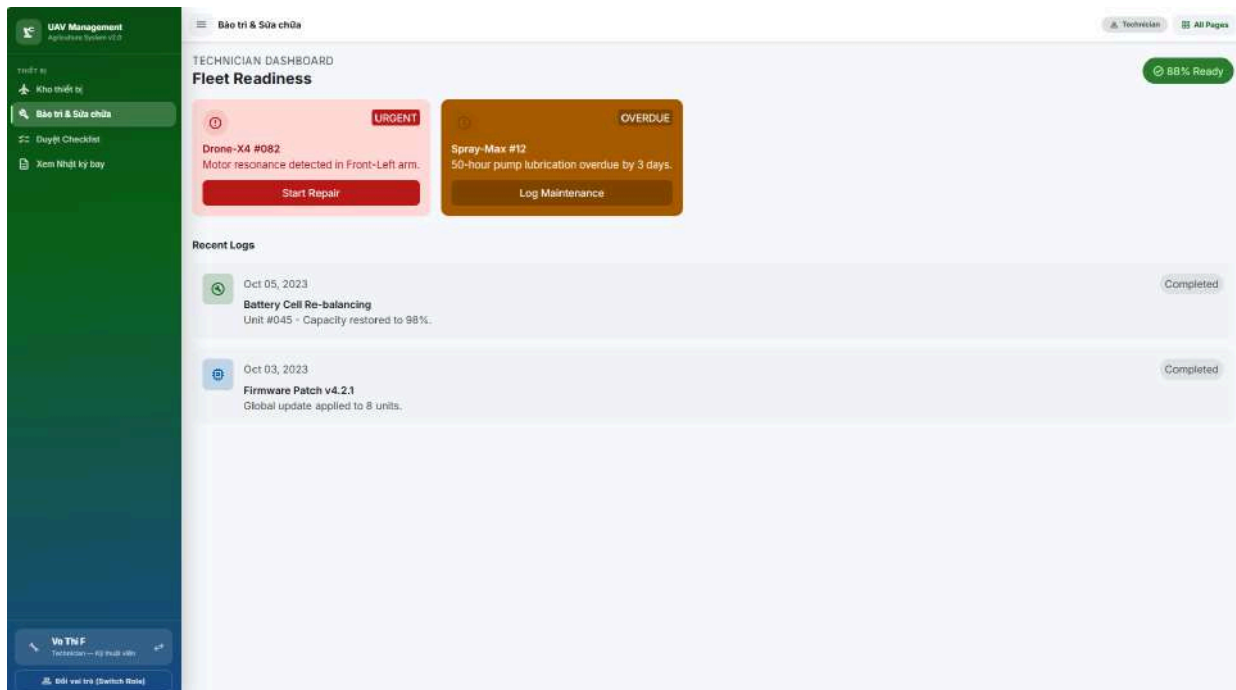
Actor: Technician

Describe: Technicians monitor maintenance schedules, repair history, and handle automated maintenance alerts to ensure the drone operates safely and reliably.

Main ingredients

- Maintenance schedule
- Repair history
- Maintenance warning
- Maintenance status
- Update





4.12 UI-12 Drone Flight Permit Verification

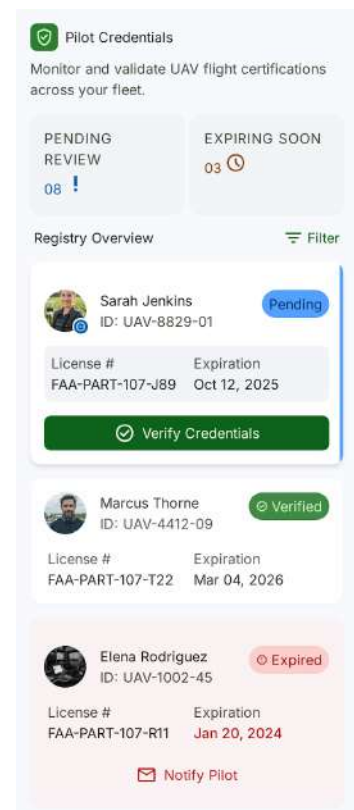
Related Use Cases: UC-11 – Verification of drone pilot's flight license

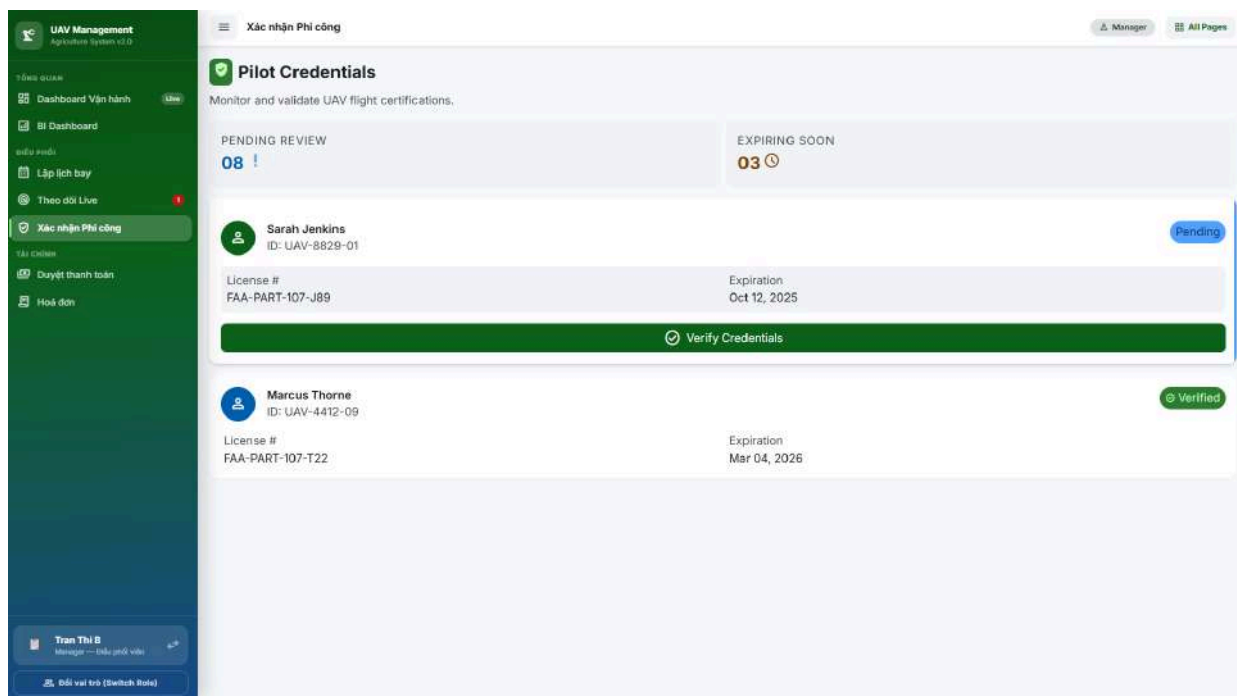
Actor: Manager

Describe: The manager verifies the validity of the drone operator's flight permit before assigning tasks to ensure compliance with applicable regulations.

Main ingredients

- Pilot information
- License information
- Expiration date
- Verification results
- Verification button





4.13 UI-13 Pre-flight Safety Checklist

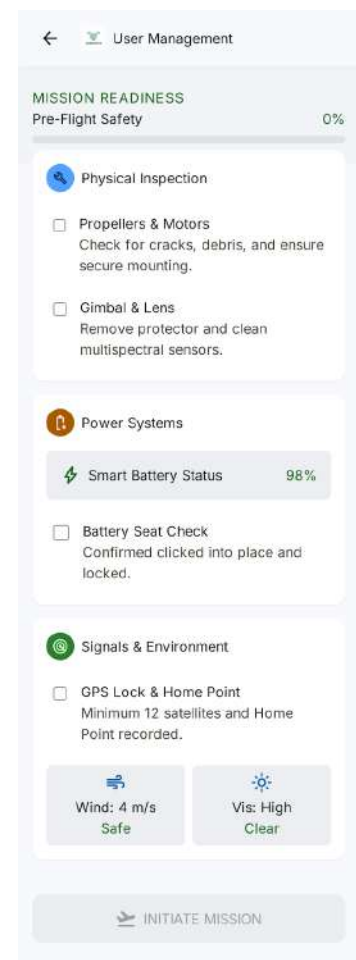
Related Use Cases: UC-12 – Complete the pre-flight safety checklist.

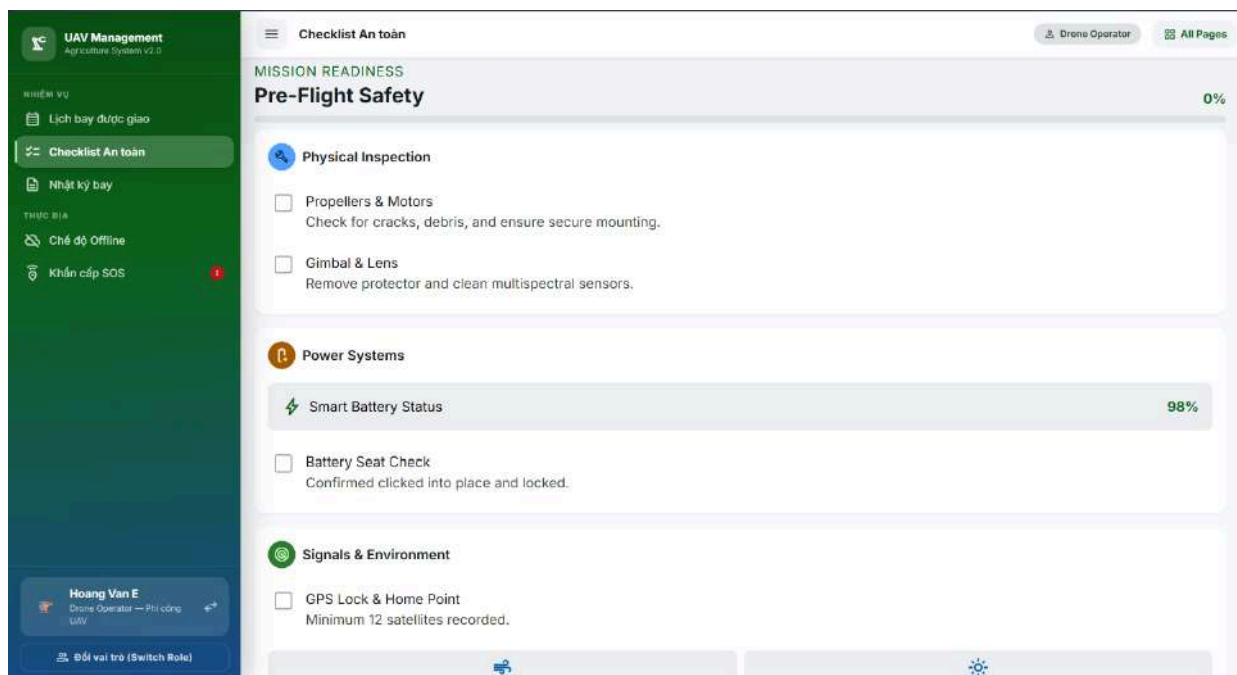
Actor: Drone Operator

Describe: Drone operators perform a safety checklist before takeoff to check the drone's condition, battery, GPS, weather conditions, and other safety items. Only after completing the checklist is the mission permitted to begin.

Main ingredients

- Mission information
- Drone inspection
- Check the battery.
- Check GPS
- Check the weather.
- Confirm Checklist





4.14 UI-14 Offline Mobile Application

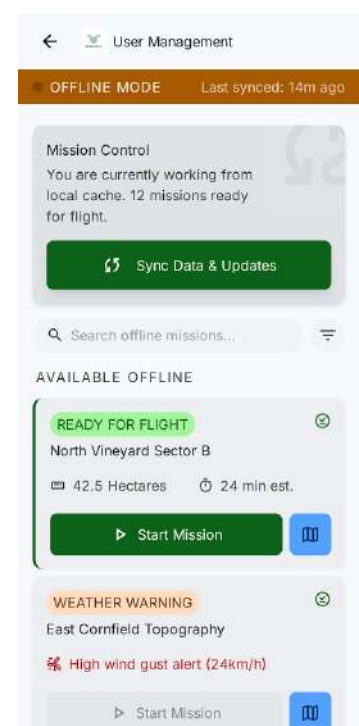
Related Use Cases: UC-13 – Operating mobile applications offline

Actor: Drone Operator

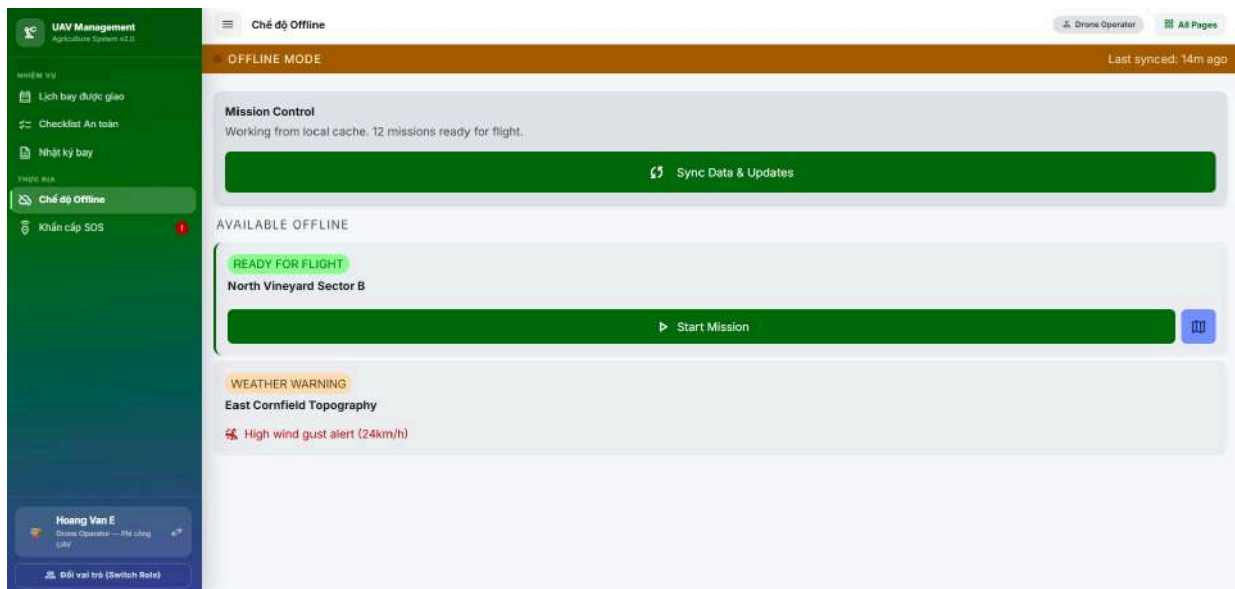
Describe: Drone Operator uses a mobile app to perform tasks even without an internet connection. Data is stored locally and automatically synchronized with the system when the connection is restored.

Main ingredients

- Task list



- Offline status
- Data synchronization
- Connection status



4.15 UI-15 Flight and Spray Log

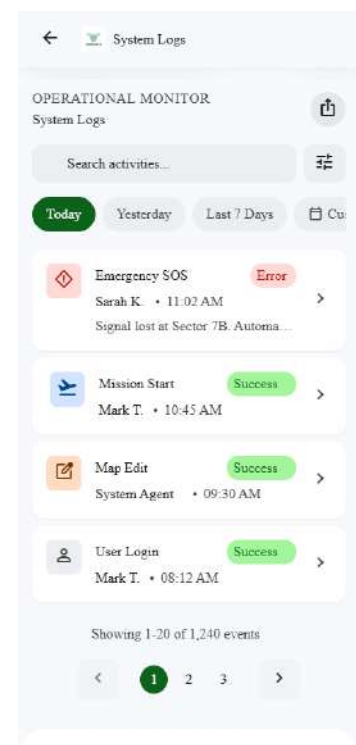
Related Use Cases: UC-14 – Data recording and flight/spray log

Actor: Drone Operator

Describe: The drone operator records information after completing the mission, including flight time, sprayed area, amount of pesticide used, and field images, to aid in monitoring and evaluating the operation.

Main ingredients

- Mission information
- Flight time
- Area sprayed
- Amount of medication to use
- Scene images
- Keep a log



UAV Management
Agriculture System V2.0

Nhật ký bay

MISSION RECORD
New Flight Log
Complete the post-flight report to sync data.

Mission Telemetry

Flight Duration: 0 min

Sprayed Area: 0.0 ha

Chemical Used: Glyphosate Ultra 450

Field Observations: Wind speed, conditions observed...

Discard | Save Mission Log

Hoàng Văn E
Drone Operator — Phi công UAV

Đổi vai trò (Switch Role)

4.16 UI-16 Emergency Alert

Related Use Cases: UC-08 – Emergency Response

Actor: Drone Operator

Describe: The drone operator sends an emergency alert when an incident occurs during the mission. The system records the incident information and current location and immediately notifies the manager for timely action.

Main ingredients

- SOS button
- Type of incident
- GPS location
- Incident description
- Image
- Processing status

User Management

Emergency SOS
Immediate broadcast to regional flight control and emergency responders.

TRIGGER ALERTS

Incident Type: Critical Hardware Failure

Latitude: 34.0522° N | Longitude: 118.2437° W

Situation Description: Describe the hazard (e.g., "Smoke detected from motor B2, landing in corn field")

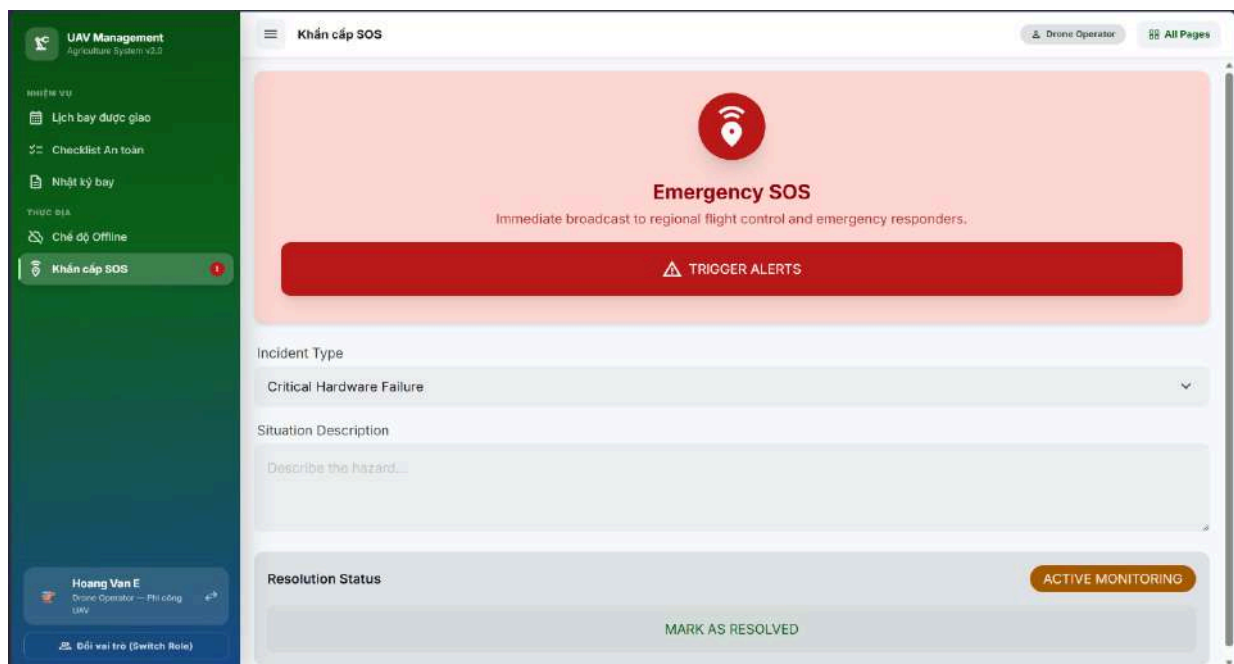
UAV-09 Last Ping

Resolution Status: ACTIVE MONITORING

Broadcast Sent: 14:02 - Signal acknowledged by ATC.

On-site Verification: Awaiting operator report or manual override.

MARK AS RESOLVED



4.17 UI-17 Order Tracking

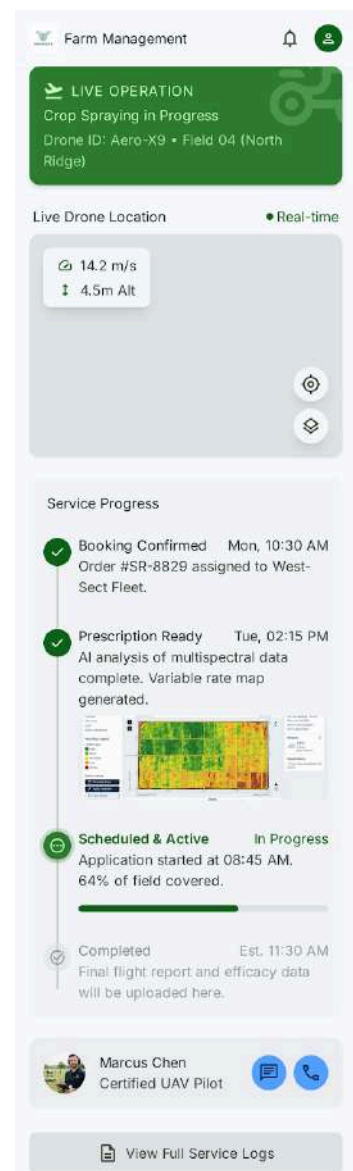
Related Use Cases: UC-15 – Visual order status tracking

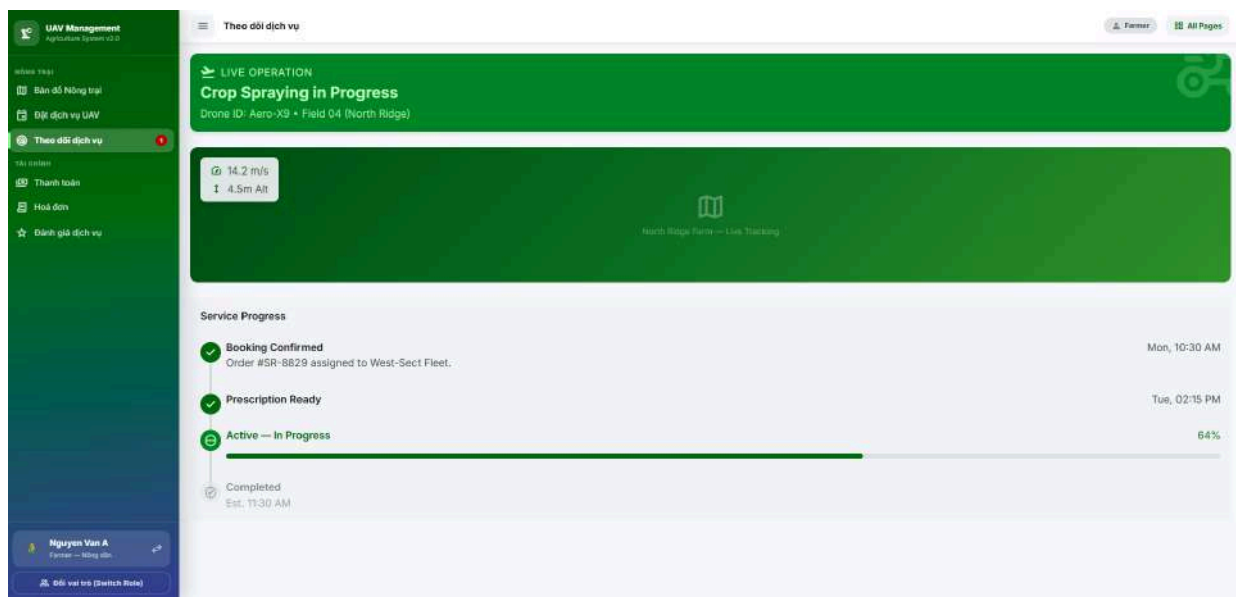
Actor: Farmer

Describe: The farmer tracks the status of service requests from the moment the request is submitted, through approval, coordination, task execution, completion, and payment.

Main ingredients

- Order process
- Current status
- Drone assigned
- Execution time
- Tracking map





4.18 UI-18 Electronic Invoice

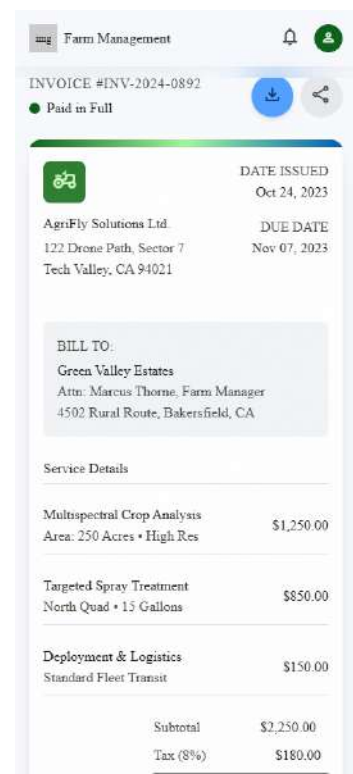
Related Use Cases: UC-16 – Invoice Management and Payment Control

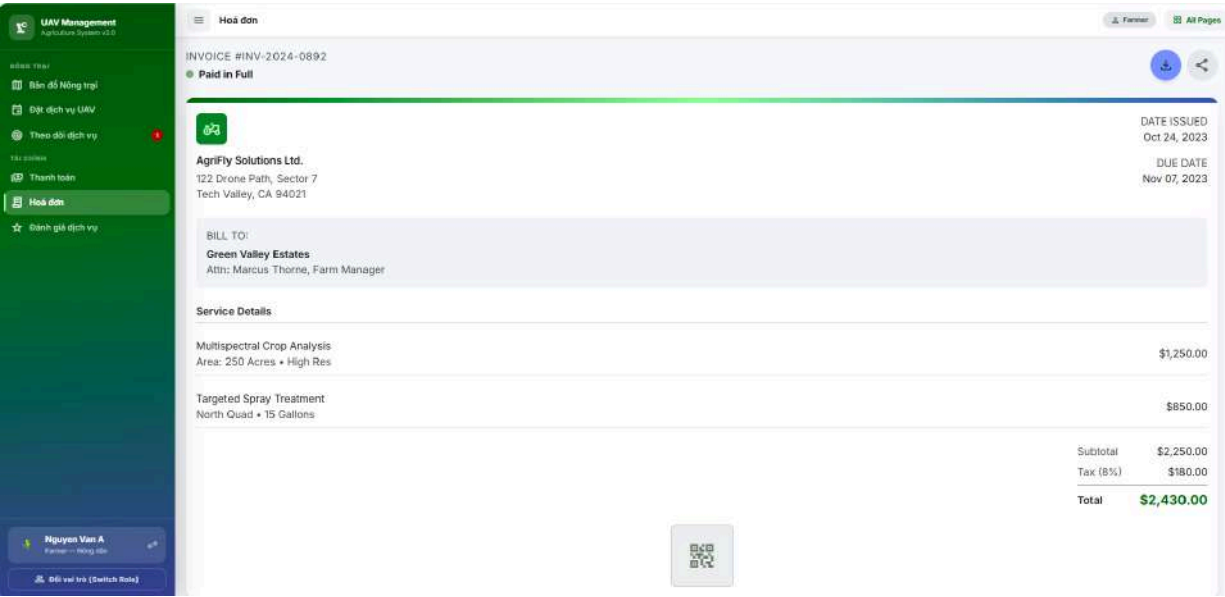
Actor: Farmer, Manager

Describe: The system displays electronic invoices after tasks are completed. Farmers can view invoice details, while Managers monitor payment status and manage service invoices.

Main ingredients

- Invoice details
- Service information
- Total amount
- Payment status
- Download PDF





4.19 UI-19 Payment Confirmation

Related Use Cases: UC-16 – Invoice Management and Payment Control

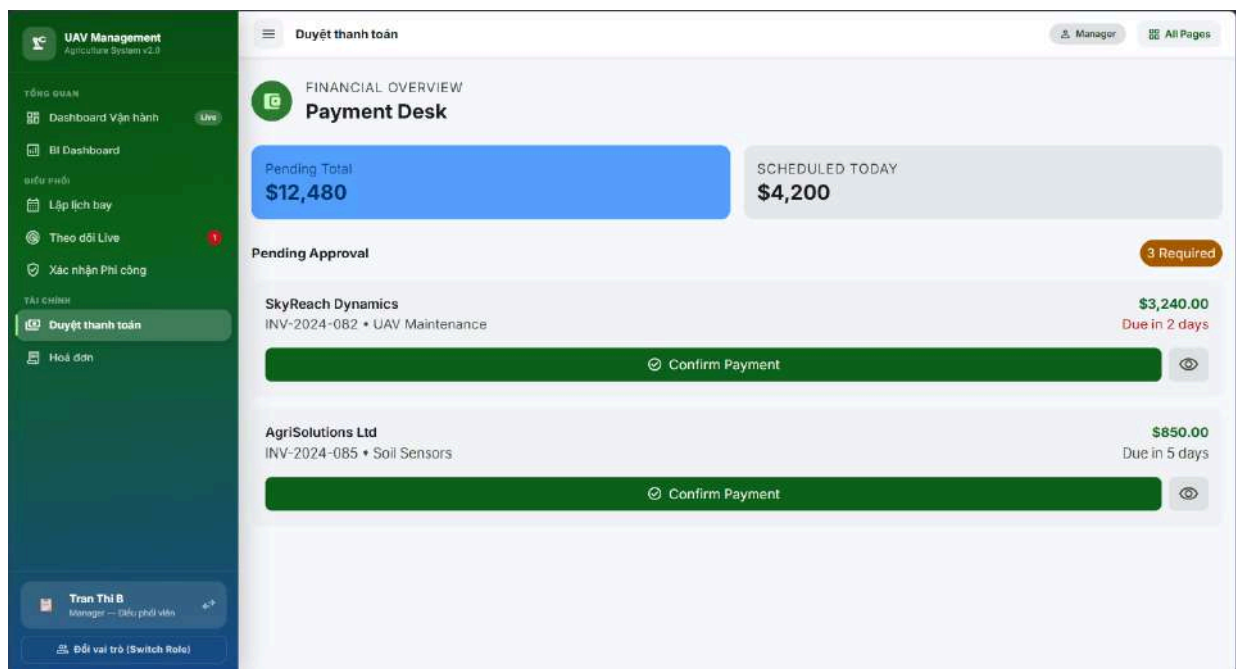
Actor: Manager

Describe: The manager confirms customer payments, updates payment status, and saves transaction history for financial management purposes.

Main ingredients

- List of invoices
- Payment details
- Confirm payment
- Payment history





4.20 UI-20 Service Review

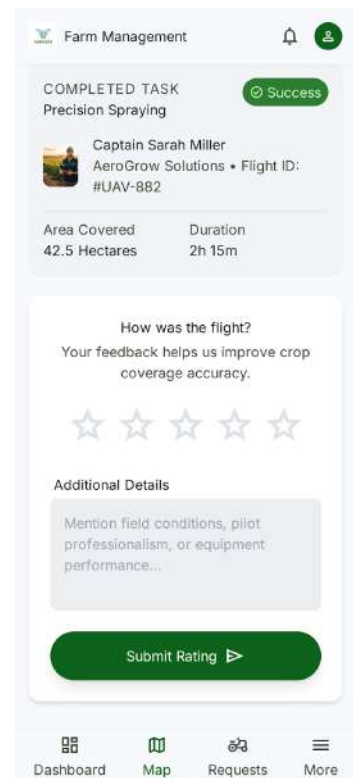
Related Use Cases: UC-17 – Service Quality Assessment Report

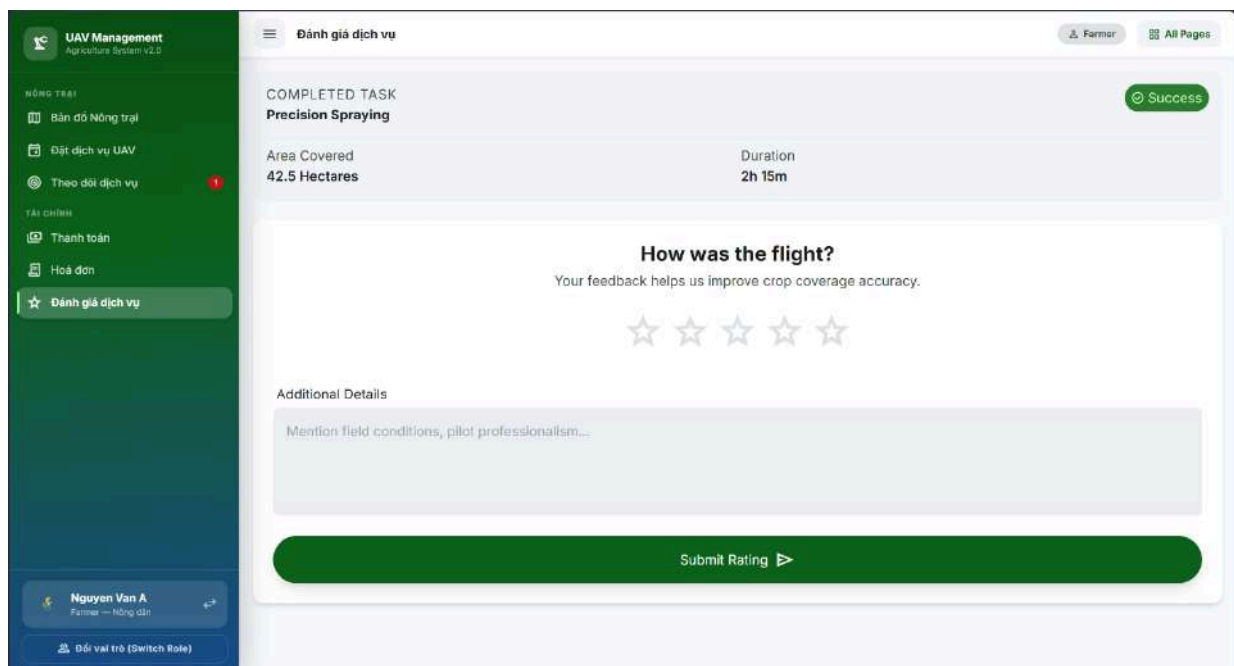
Actor: Farmer

Describe: After the service is completed, the Farmer evaluates the service quality through star ratings and comments, helping the system record feedback and improve service quality.

Main ingredients

- Star rating
- Comment
- Service information
- Submit a review





4.21 UI-21 Dashboard report

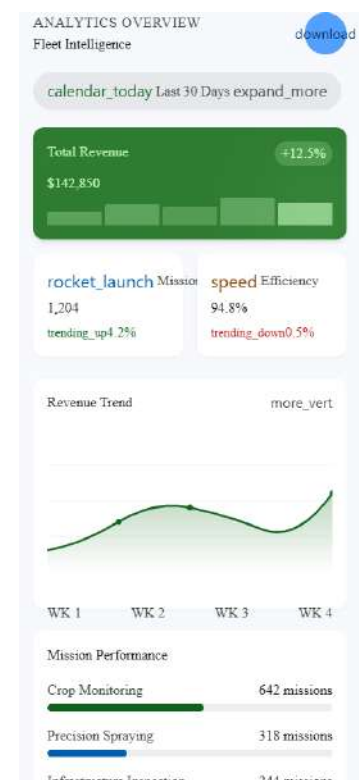
Related Use Cases: UC-18 – Revenue Report Compilation and Data Export

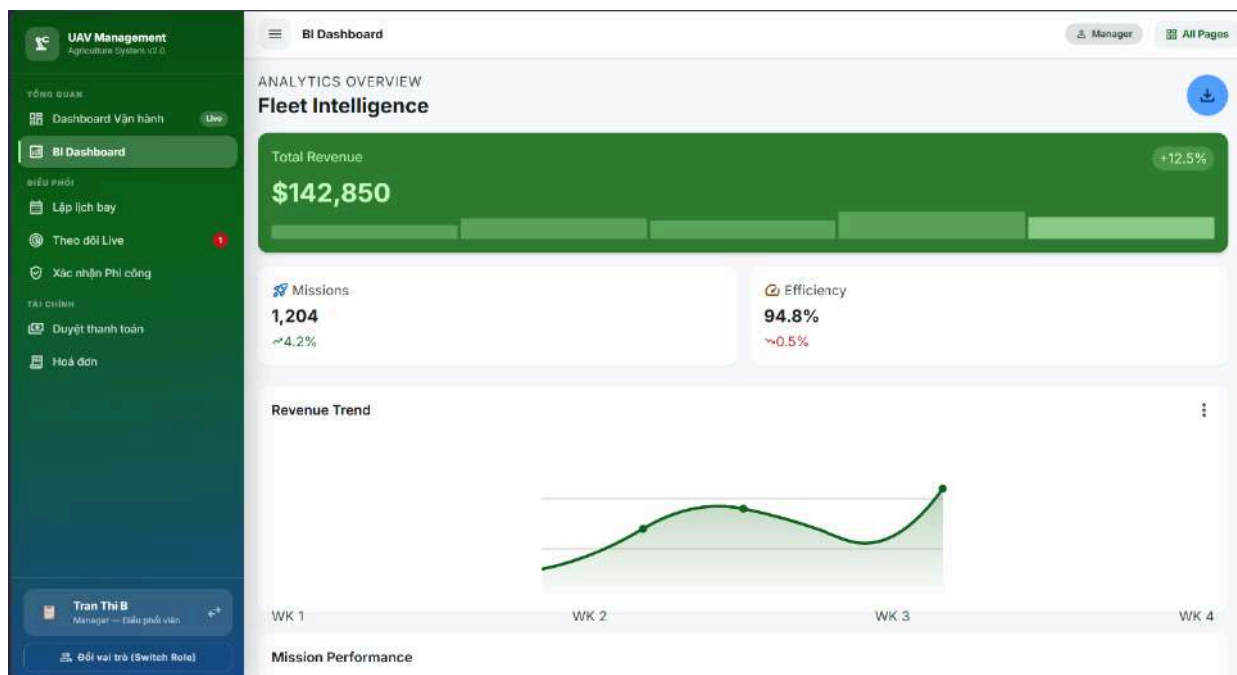
Actor: Manager

Describe: Managers monitor aggregated reports on revenue, mission count, drone performance, and operational metrics. Reports can be filtered by time period and exported to PDF or Excel formats.

Main ingredients

- Time filter
- KPI Card
- Revenue report
- Task statistics
- Drone Performance
- Export PDF
- Export to Excel





UI ID	Screen name	Actor	Use Case related
UI-01	Account management	Administrator	UC-01 – Centralized Account Management
UI-02	System logs	Administrator	UC-02 – Operation monitoring and fault lookup
UI-03	Login screen	All users	UC-03 – Login and security authentication
UI-04	Land map and record management	Farmer	UC-04 – Land Map and Record Management
UI-05	Schedule a service	Farmer	UC-05 – Schedule agricultural spraying services
UI-06	Task coordination	Manager	UC-06 – Flight schedule coordination and resource monitoring
UI-07	Monitoring dashboard	Manager	UC-06 – Flight schedule coordination and resource monitoring
UI-08	Electronic prescription	Agronomist	UC-07 – Drug Request and Prescription Review
UI-09	Testing for banned substances	Agronomist	UC-07 – Drug Request and Prescription Review
UI-10	Device catalog	Technician	UC-09 – Drone and Battery Catalog

	management		Management
UI-11	Maintenance management	Technician	UC-10 – Automated Maintenance Management and Alert System
UI-12	Verify drone flight permit.	Manager	UC-11 – Verification of drone pilot's flight license
UI-13	Pre-flight safety checklist	Drone Operator	UC-12 – Complete the pre-flight safety checklist.
UI-14	Offline mobile app	Drone Operator	UC-13 – Operating mobile applications offline
UI-15	Flight and spray log	Drone Operator	UC-14 – Data recording and flight/spray log
UI-16	Emergency alert	Drone Operator	UC-08 – Emergency Response
UI-17	Track your order	Farmer	UC-15 – Visual order status tracking
UI-18	Electronic invoices	Farmer, Manager	UC-16 – Invoice Management and Payment Control
UI-19	Confirm payment	Manager	UC-16 – Invoice Management and Payment Control
UI-20	Service evaluation	Farmer	UC-17 – Service Quality Assessment Report
UI-21	Dashboard reports	Manager	UC-18 – Revenue Report Compilation and Data Export

5. UI Screen Specification

This section describes the system's main interface screens. **Agriculture UAV Management System** This includes the intended use, target users, relevant use cases, interface components, and main functions of each screen. The interfaces are designed to be intuitive, easy to use, and appropriate for each role to support users in performing tasks efficiently.

5.1 UI-01 – Account Management Screen

Target users: Administrator

Use Case: UC-01 – Centralized Account Management

Purpose: This allows the administrator to manage user accounts, assign permissions by role, enable or disable accounts, and reset passwords when needed.

Interface components

- Account list
- Search bar
- Role filter
- Account information
- Add account button
- Edit button
- Permissions button
- Activate/Deactivate Button
- Reset Password button

Function

- Add a new account.
- Update your account information.
- User permission management.
- Activate or deactivate your account.
- Reset your password.
- Search and filter accounts.

5.2 UI-02 – System Log Screen

Target users: Administrator

Use Case: UC-02 – Operation monitoring and fault lookup

Purpose: This allows the administrator to monitor system activities, check logs, and look up errors to support administration and ensure system security.

Interface components

- System log
- User filter
- Active type filter
- Time filter
- Error details
- Search bar
- Export log button

Function

- Track user activity.
- Look up system errors.
- Filter logs by time or user.
- Search the logs.
- Generate logs for inspection purposes.

5.3 UI-03 – Login Screen

Target users:All users

Use Case: UC-03 – Login and security authentication

Purpose: Allows users to log into the system using their assigned accounts to access functions appropriate to their roles.

Interface components

- System logo
- Email input field
- Password input field
- Remember me
- Login button
- Forgot Password Link
- Error message

Function

- Check your login information.
- User authentication.
- Switch to the role-based interface.
- Display a message when login fails.

5.4 UI-04 – Land Map and Record Management Screen

Target users:Farmer

Use Case: UC-04 – Land Map and Record Management

Purpose: This allows farmers to manage information about their land plots, including location, area, crop types, and farming history.

Interface components

- Map
- List of land plots
- Land plot information
- Type of crop
- GPS coordinates
- Edit button
- Save button

Function

- Add or update land plot information.
- Display your location on the map.
- Crop information management.
- Save land records and information.

5.5 UI-05 – Service Scheduling Screen

Target users:Farmer

Use Case: UC-05 – Schedule agricultural spraying services

Purpose: This allows farmers to request drone spraying services by providing information about their farming area and service needs.

Interface components

- Customer information
- Choose a plot of land
- Type of crop
- Describe the pest and disease situation.
- Desired date
- Download image
- Submit Request button
- Cancel button

Function

- Enter the service request information.
- Verify the input data.
- Send a request to the system.
- Save the request with a pending status.

5.6 UI-06 – Task Coordination Screen

Target users: Manager

Use Case:UC-06 – Flight schedule coordination and resource monitoring

Purpose: This allows managers to assign drones, drone operators, and task schedules, while also detecting and resolving resource conflicts.

Interface components

- List of pending tasks
- List of available drones
- Drone Operator List
- Implementation schedule
- Conflict alert
- Assignment Button

Function

- Select Drone and Drone Operator.
- Check for calendar conflicts.
- Assign tasks.
- Update the mission status.

5.7 UI-07 – Monitoring Dashboard

Target users: Manager

Use Case:UC-06 – Flight schedule coordination and resource monitoring

Purpose: This display provides an overview of the drone's operational status, mission progress, and resource usage, assisting managers in monitoring the system in real time.

Interface components

- KPI Card
- Drone status
- Task progress
- Resource status
- Notification
- Monitoring map

Function

- Track task progress.
- Monitor drone status.
- Monitor resource usage.
- Display important notifications.
- Quick access to management functions.

5.8 UI-08 – Electronic Prescription Display Screen

Target users: Agronomist

Use Case:UC-07 – Drug Request and Prescription Review

Purpose:This allows the Agronomist to review the service request, assess the crop condition, and create a suitable electronic prescription before moving on to the task coordination step.

Interface components

- Information required
- Plant information
- Pest and disease assessment
- Recommended pesticides
- Dosage
- Approve button
- Reject button

Function

- View service requirements.
- Assess the condition of the crops.
- Create electronic prescriptions.
- Approve or reject the request.

- Save the prescription to the system.

5.9 UI-09 – Screen for Checking Prohibited Substances

Target users: Agronomist

Use Case: UC-07 – Drug Request and Prescription Review

Purpose: Agronomist assists in checking proposed pesticides for banned active ingredients or those that do not meet current regulations before approving prescriptions.

Interface components

- Active ingredient search box
- List of banned substances
- Test results
- Warning notification
- Suggested alternatives

Function

- Look up the active ingredient.
- Check the list of banned substances.
- Display a violation warning.
- Suggest a suitable alternative medication.
- Assisting in completing prescriptions.

5.10 UI-10 – Device Catalog Management Screen

Target users: Technician

Use Case: UC-09 – Drone and Battery Catalog Management

Purpose: This allows technicians to manage drone and battery information, update device status, and monitor readiness for flight missions.

Interface components

- Drone List
- Drone Information
- Battery Status
- Number of flight hours
- Search bar
- State filter
- Update button

Function

- Add a new drone.
- Drone information update.

- Manage battery information.
- Track your flight hours.
- Update the drone's operational status.
- Search and filter devices.

5.11 UI-11 – Maintenance Management Screen

Target users: Technician

Use Case: UC-10 – Automated Maintenance Management and Alert System

Purpose: This allows technicians to track maintenance schedules, manage repair history, and receive automatic maintenance alerts to ensure the drone is always ready for operation.

Interface components

- Maintenance schedule
- Repair history
- Maintenance warning
- Maintenance status
- Update button

Function

- Check your drone's maintenance schedule.
- Update maintenance results.
- Track your repair history.
- Receive automatic maintenance alerts.
- Update status after maintenance.

5.12 UI-12 – Drone Flight Permit Authentication Screen

Target users: Manager

Use Case: UC-11 – Verification of drone pilot's flight license

Purpose: Allows the Manager to verify the validity of the flight permit before assigning the Drone Operator to perform a mission.

Interface components

- Drone Operator Information
- Flight permit information
- Date issued
- Expiration date
- Verification results
- Verification button

Function

- Look up your flight permit.
- Check the license expiration date.
- Verify the conditions for performing the task.
- Display a warning if the license is invalid.

5.13 UI-13 – Pre-flight safety checklist screen

Target users: Drone Operator

Use Case: UC-12 – Complete the pre-flight safety checklist.

Purpose: This allows the drone operator to perform pre-flight safety checks to ensure the drone and operating conditions meet requirements before takeoff.

Interface components

- Mission information
- Drone inspection
- Check the battery.
- Check GPS
- Check the weather.
- Checklist Confirmation Button

Function

- Complete the checklist before your flight.
- Confirm drone status.
- Check the weather conditions.
- Save the test results.
- Allow tasks to begin after completing the checklist.

5.14 UI-14 – Offline Mobile App Screen

Target users: Drone Operator

Use Case: UC-13 – Operating mobile applications offline

Purpose: This allows the drone operator to continue its mission even when the internet connection is lost and to synchronize data with the system when the connection is restored.

Interface components

- Task list
- Offline status
- Local data storage
- Data Sync button
- Connection status

Function

- View missions offline.
- Record data locally.
- Data synchronization is available when an internet connection is present.
- Display the synchronization status.

5.15 UI-15 – Flight Log and Spray Monitor

Target users: Drone Operator

Use Case: UC-14 – Data recording and flight/spray log

Purpose: This allows the drone operator to record all information after completing the flight and spraying mission, for the purpose of monitoring and storing operational data.

Interface components

- Mission information
- Flight time
- Area sprayed
- Amount of medication to use
- Scene images
- Save Log button

Function

- Record flight entries in the flight log.
- Keep a log of pesticide spraying.
- Update the medication dosage.
- Attached are photos of the scene.
- Save the data to the system.

5.16 UI-16 – Emergency Alert Screen

Target users: Drone Operator

Use Case: UC-08 – Emergency Response

Purpose: This allows the drone operator to send emergency alerts when incidents occur during a mission, enabling the manager to take timely action.

Interface components

- SOS button
- Type of incident
- GPS location
- Incident description
- Upload photos of the scene
- Processing status

Function

- Send an emergency signal.
- Send your current location.
- Report an incident.
- Attached is an image.
- Monitor the troubleshooting status.

5.17 UI-17 – Order Tracking Screen

Target users:Farmer

Use Case: UC-15 – Visual order status tracking

Purpose:This allows the Farmer to track the entire process of handling service requests, from request submission to completion.

Interface components

- Order process
- Current status
- Drone assigned
- Execution time
- Tracking map

Function

- Check your order status.
- Monitor the progress of implementation.
- View the task schedule.
- Track the location on the map.

5.18 UI-18 – Electronic Invoice Screen

Target users: Farmer, Manager

Use Case:UC-16 – Invoice Management and Payment Control

Purpose:This allows farmers to view electronic invoices after the service is completed and assists managers in managing invoices within the system.

Interface components

- Invoice details
- Service information
- Total amount
- Payment status
- Download PDF button

Function

- Display electronic invoices.

- Track your payment status.
- Download the invoice as a PDF.
- Manage the list of invoices.

5.19 UI-19 – Payment Confirmation Screen

Target users: Manager

Use Case: UC-16 – Invoice Management and Payment Control

Purpose: Allows the Manager to confirm customer payments and update the payment status of invoices.

Interface components

- List of invoices
- Payment details
- Payment status
- Payment history
- Payment Confirmation Button

Function

- View payment information.
- Confirm payment.
- Update invoice status.
- Save transaction history.

5.20 UI-20 – Service Review Screen

Target users: Farmer

Use Case: UC-17 – Service Quality Assessment Report

Purpose: This allows farmers to evaluate the quality of service after the task is completed, providing feedback to the system and the service provider.

Interface components

- Star rating
- Comment
- Service information
- Submit Review button

Function

- Evaluating service quality.
- Enter a comment.
- Submit feedback.
- Save the review to the system.

5.21 UI-21 – Reporting Dashboard Screen

Target users: Manager

Use Case: UC-18 – Revenue Report Compilation and Data Export

Purpose: This allows managers to monitor aggregated reports on revenue, missions, drone performance, and operational metrics, and to export data for management purposes.

Interface components

- Time filter
- KPI Card
- Revenue report
- Task statistics
- Drone Performance
- Statistical chart
- PDF Export Button
- Export Excel button

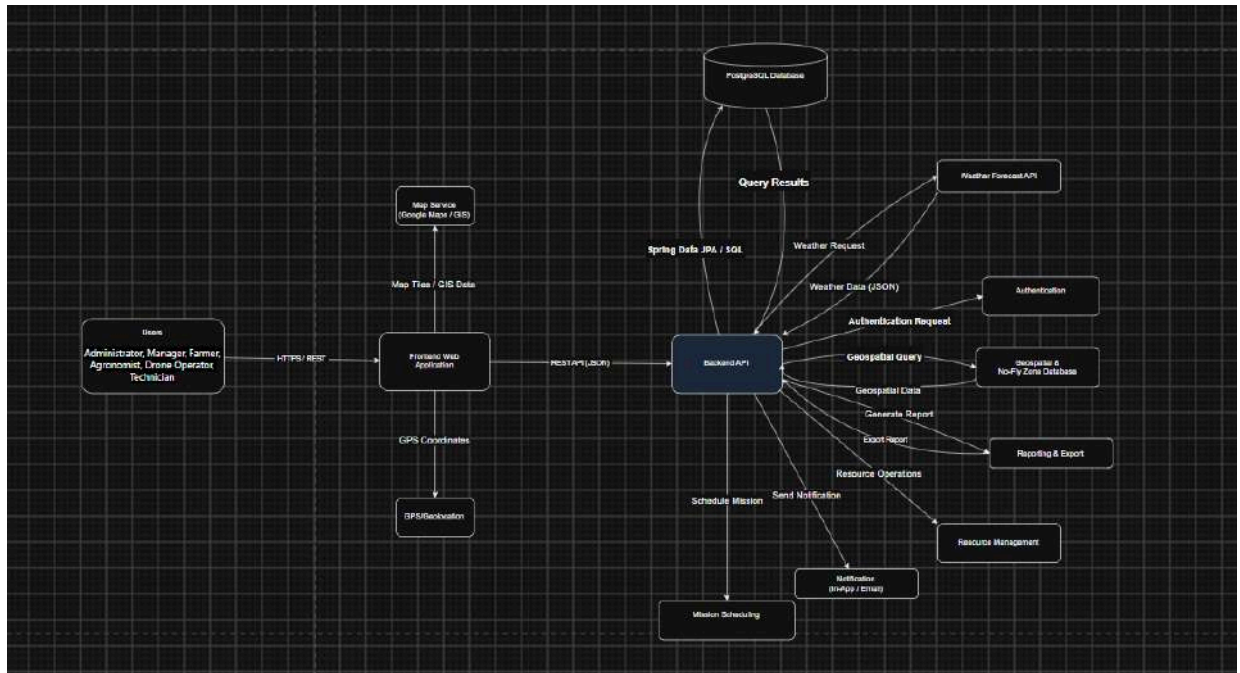
Function

- View reports by time period.
- Track revenue.
- Task statistics.
- Monitor drone performance.
- Export the report as a PDF.
- Export the report as an Excel spreadsheet.

6. System Architecture

System **Agriculture UAV – Precision Spraying Management Platform** designed according to architecture **3-Layer (Modular Monolithic Architecture)** This architecture helps separate the user interface, business logic, and data storage, thereby increasing maintainability, scalability, and testability. Each layer handles a separate set of functions and communicates with each other through internal APIs.

For a course module project with a development time of approximately 10 weeks, this architecture reduces complexity compared to Microservices while still ensuring scalability when new functionalities need to be added in the future.



Layer Responsibilities

Layer	Responsibility
Presentation Layer	It provides a user interface through a web application. It receives user input, displays data, and sends requests to the business logic layer via a REST API.
Application Layer	Perform system business processes such as user authentication, authorization, service request management, task coordination and emergency incident handling, drone management, log recording, and report generation. Simultaneously verify input data and apply Business Rules..
Data Layer	Store and manage system data using PostgreSQL. Perform data retrieval via Spring Data JPA, ensuring data integrity, maintaining audit logs, and supporting data backups.

Data Flow Overview

1. Users access the system through a web browser.
2. React interfaces send requests to Spring Boot's REST APIs.
3. Spring Boot authenticates users using JWT and performs rights and access checks

(RBAC).

4. The system processes business requests on demand, applying Business Rules and Validation.
5. Data is read or written to the PostgreSQL database via Spring Data JPA.
6. The results are returned to the interface for display to the user.

7. Architecture Trade-off Analysis

To select the appropriate architecture for the system.**Agriculture UAV – Precision Spraying Management Platform**The group compared two popular options:**3-Layer Modular Monolithic Architecture** and **Microservices Architecture**.

After evaluating based on project scale, development time, and team resources, the architecture**3-Layer Modular Monolithic**It was chosen because it effectively meets current requirements while reducing complexity in the development and deployment process.

Criteria	Option 1: 3-Layer Monolithic	Option 2: Microservices	Selected
Complexity	Low, easy to design and develop.	Tall, requires managing many independent services.	3-Layer
Suitable for a 10-week project.	Very suitable	Unsuitable due to short implementation time	3-Layer
Scalability	Medium size, expandable via module.	Very high, expanding each service independently.	Microservices
Communication performance	Modules that are called directly within the same application should be faster.	Communication via API between services should have some latency.	3-Layer
Development costs	Short	High price due to the need for many services and infrastructure.	3-Layer

Implementation costs	Simply put, all you need is an application and a database.	Complex, requires API Gateway, Service Discovery, Containers, etc.	3-Layer
Maintainability	Easy to maintain if organized in a clear modular structure.	Good for large systems but more complex for smaller projects.	3-Layer
Suitability for project scale	Suitable for the current student group and functional scope.	Suitable for large enterprise systems	3-Layer

7.1 Reasons for choosing 3-Layer architecture

Selection group **3-Layer Modular Monolithic Architecture** for the following reasons:

- Suitable for the scale of the course project and the development timeframe of approximately **10 weeks**.
- Reduces complexity in development, testing, and deployment compared to Microservices architecture.
- Allows for a clear separation between **Presentation Layer**, **Application Layer** and **Data Layer** This makes the source code easier to maintain and extend.
- It supports development across various business modules such as User Management, Task Coordination and Emergency Response, Drone Management, Flight Logs, and Reporting, all while operating within a unified application..
- Easy to deploy on a simple server or cloud environment, suitable for project demonstration and evaluation goals.

7.2 Future Expansion Directions

Although a 3-layer architecture was chosen for the current version, the system is still designed to be modular for future expansion. As the number of users or functionalities increases, modules such as **Mission Management**, **Reporting**, or **Notification Service** They can be separated into independent microservices without changing the entire system.

Conclude

After analyzing criteria such as complexity, scalability, development costs, and suitability for the project scope, the team decided to select... **3-Layer Modular Monolithic Architecture** This architecture effectively meets the current requirements of the Agriculture UAV system, while also providing a favorable foundation for future maintenance and expansion.

8. Technology Stack

System **Agriculture UAV – Precision Spraying Management Platform** using technologies that are compatible with the architecture. **3-Layer Modular Monolithic** The technologies were selected to ensure rapid development, ease of maintenance, and fulfillment of project requirements within the given timeframe.

Component	Proposed technology	Reason
UI Prototype	Figma	The wireframe and prototype design is intuitive, easy to share, and facilitates collaboration among team members.
Diagram Modeling	Draw.io (diagrams.net), Lucidchart	Create DFDs, Activity Diagrams, ERDs, and system architecture diagrams.
Frontend	React.js	Build a modern user interface that supports component-based development and is easily extensible.
Backend	Java Spring Boot	Develop RESTful APIs, handle business logic, integrate security, and ensure ease of maintenance.
Database	PostgreSQL	A robust relational database management system that supports data integrity and is compatible with management systems.
Authentication	JWT (JSON Web Token)	Secure user authentication and login session management using token-based methods.
Password Encryption	BCrypt	Encrypting passwords before saving them to the database enhances security.
ORM Framework	Spring Data JPA (Hibernate)	Simplify database operations and object mapping to data tables.
API	RESTful API	Standard communication interface between frontend and backend, easy to integrate and expand.
Version Control	Git & GitHub	Manage source code, track change history, and support teamwork.
Development Environment	Visual Studio Code, IntelliJ IDEA	We support both frontend and backend development with a variety of programming tools.

Deployment	Local Server / Cloud VM	Implement the system for the purpose of project demonstration and testing.
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Reasons for choosing

These technologies were selected because they best meet the project requirements:

- React.js helps build modern, reusable interfaces and facilitates component-based development.
- Spring Boot supports rapid development of RESTful APIs and integrates well with Spring Security and JWT for implementing authentication and authorization mechanisms.
- PostgreSQL is compatible with the system's relational data model, ensuring data integrity and stable processing capabilities.
- Spring Data JPA (Hibernate) helps reduce the amount of SQL query code, increasing maintainability and development capabilities.
- GitHub supports effective version management and collaboration among team members.
- Figma and Draw.io help design interfaces and system models intuitively, making analysis and communication easier.

9. Physical Data Design

Physical Data Design describes the main data tables used in the system. **Agriculture UAV** The database is designed using a relational model to ensure data integrity, reduce redundancy, and effectively support user management, task coordination, drone management, and reporting functions.

Table	Purpose
Users	Manage user account information, including login details, roles, and account status.
Fields	Record information about farmers' land plots, such as location, area, and type of crop.
Service_Requests	Manage requests for pesticide spraying services submitted by farmers.
Missions	Manage information on spraying missions, assign personnel, drones, and mission status.
Prescriptions	Electronic prescriptions are saved for each task.
Drones	Manage drone information, operating status, and technical specifications.

Batteries	Manage drone battery information, including charge cycles and usage status.
Operation_Logs	Record operational information during the mission, such as flight time and battery consumption.
Spray_Logs	Record information about the spraying process, such as the area sprayed, the amount of pesticide used, and the time of application.
Maintenance_Records	Manage the maintenance history, costs, and results of drone maintenance.
Operator_Licenses	Manage information on drone operator certificates and licenses.
Invoices	Manage invoices, payment methods, and payment status for each task.
Emergency_Reports	Record any incidents that occur during the execution of the task and their processing status.
Reviews	Save farmers' reviews and feedback after the service is completed.
System_Audit_Logs	Recording user and system activity history for inspection, monitoring, and tracing purposes.
No_Fly_Zones	Record the boundaries of no-fly/restricted airspace to facilitate overlap checks when creating service requests or coordinating missions.
Banned_Chemicals	This list of banned plant protection active ingredients is used for automatic reference when prescribing pesticides.
Preflight_Checklists	Record the results of the mandatory safety checks performed on the drone operator before each flight.

10. Security Design

System security **Agriculture UAV** Designed to protect user data, operational data (flight missions, prescriptions, spray logs), and ensure compliance with flight safety and pesticide regulations. The security design employs four layers: **Authentication, Authorization, Data Protection, Auditability**.

10.1 Authentication & Session Management

- Users log in using email/password; the password is hashed using **BCrypt** Before saving, do not save plaintext.

- After successful verification, the system grants **JWT (JSON Web Token)** There is an expiration period for managing login sessions.
- Apply the mechanism **session inactivity timeout** Automatically logs out if the user is inactive for a specified period, reducing the risk of session hijacking when the device is left unattended.
- Accounts are only activated after OTP verification via email, preventing the creation of spam/fake accounts within the system.

10.2 Authorization (RBAC)

The system applies the model **Role-Based Access Control** with 6 roles: Administrator, Manager, Farmer, Agronomist, Drone Operator, Technician.

Role	Access scope
Administrator	Full control: manage accounts, assign permissions, view system-wide audit logs.
Manager	Coordinate tasks, assign Drone/Operator duties, and view reports.
Farmer	Only access your own service requests and data.
Agronomist	Assess the request and create an electronic prescription.
Drone Operator	Update the flight/spray log for the assigned mission.
Technician	Manage devices, batteries, and maintenance schedules.

Each API endpoint is subject to role checks at the Backend (Spring Security) level, not just hidden/visible on the Frontend interface — preventing users from directly calling the API to bypass UI controls.

10.3 Data Protection

- The entire Client-Server communication uses **HTTPS/TLS**.
- Sensitive data (contact information, land parcel coordinates, flight location logs) is only displayed to the respective authorized roles.
- Input validation and cleaning are performed on both the frontend and backend to minimize errors. **SQL Injection** and **Cross-Site Scripting (XSS)**.

10.4 Threat & Mitigation Matrix

ID	Threat	Asset affected	Likelihood	Impact	Mitigation
T-01	Unauthorized access	Mission data, user data	Medium	Data leakage	Authentication + RBAC theo role

T-02	Wrong drone/operator assignment	Mission operation	Medium	Operational error, delay	Business rule validation before assignment (checking readiness status).
T-03	Tampering operation/spray log	Operation logs, Spray logs	Low	Loss of traceability, legal risk	Append-only audit logs have restricted permissions (only operators can write to them, they cannot be modified after submission).
T-04	Leakage of location/airspace data	Field location, flight path	Medium	Safety/privacy risk	Access is restricted by role; land plot coordinates are not to be made public.
T-05	Loss of telemetry/log data	Operation logs, Spray logs	Low	Missing reports, lack of reference material.	Regular backup plan (item 12)
T-06	Brute-force / credential stuffing	User accounts	Medium	Account takeover	Limit the number of failed login attempts, OTP verification during registration.
T-07	SQL Injection / XSS	The entire database	Low	Data corruption, leakage	Input validation using Prepared Statement/JPA (no raw SQL string concatenation).
T-08	Violation of no-fly zone / prohibited chemicals	Compliance, business reputation	Medium	Legal risks	Check the list of prohibited chemicals when creating prescriptions (business rule).

10.5 Auditability

All important operations (login, request approval, task assignment, status changes, administrative actions) are recorded. **System_Audit_Logs** Only administrators have access to this, for incident investigation and compliance audit purposes.

11. QA and Test Strategy

The testing strategy applies functional testing based on the use cases specified in Deliverable 2 & 3, combined with negative testing for critical business processes such as drone assignment, prescription dispensing, and billing.

Test Case ID	Related UC	Test Objective	Test Data	Expected Result
TC-01	UC-03	Login with valid account	Valid email/password	User accesses dashboard
TC-02	UC-03	Login with invalid password	Wrong password	System shows error, access denied
TC-03	UC-01	Admin creates new staff account	Valid role & info	Account created with correct role
TC-04	UC-02	Admin views system audit log	Existing log entries	Log list displayed with timestamp, actor, action
TC-05	UC-04	Farmer creates field profile	Valid field boundary & area	Field is saved and shown on map
TC-06	UC-05	Create valid mission/service request	Valid mission data	Request is created as Pending
TC-07	UC-05	Create request without location	Missing location	System shows validation error
TC-08	UC-07	Agronomist approves pending request	Pending request	Request becomes Approved.
TC-09	UC-07	Agronomist prescribes banned chemical	Chemical in blacklist	System blocks prescription, shows warning
TC-10	UC-06	Assign available	Available	Assignment is created,

		drone & operator	drone/operator	status → Scheduled
TC-11	UC-06	Assign unavailable drone	Drone status = unavailable	System rejects assignment
TC-12	UC-06	Detect scheduling conflict	Operator already assigned in same time slot	System shows conflict warning
TC-13	UC-11	Verify operator flight license before assignment	Expired license	System blocks assignment, shows alert
TC-14	UC-08	Report emergency incident during mission	Incident data (mission ID, description)	Incident logged, mission status → Emergency
TC-15	UC-09	Technician registers new drone	Valid drone specs	Drone added to fleet list
TC-16	UC-10	System triggers maintenance alert	Battery cycle exceeds threshold	Alert generated, drone marked "Needs Maintenance"
TC-17	UC-12	Operator completes pre-flight checklist	All checklist items checked	Mission allowed to proceed
TC-18	UC-12	Operator skips required checklist item	Missing safety item	System blocks mission start
TC-19	UC-14	Operator submits operation/spray log	Valid operation & spray data	Log saved, mission status → Completed
TC-20	UC-15	Farmer tracks request/mission	Existing request ID	Current status displayed correctly

		status		
TC-21	UC-16	Generate invoice after mission completion	Completed mission	Invoice created with correct amount
TC-22	UC-16	Confirm payment for invoice	Valid payment confirmation	Invoice status → Paid
TC-23	UC-17	Farmer submits service review	Rating + comment	Review saved and linked to mission
TC-24	UC-18	Manager generates revenue report	Date range filter	Report data matches completed missions in range

12. Deployment and Operations Plan

Item	Description
Deployment model	The web-based application is implemented using a three-tier model: Frontend (React.js) and Backend (Spring Boot), packaged and run on a Cloud VM (e.g., AWS EC2/DigitalOcean Droplet) or the school's local server for demonstration and testing purposes within the ITA301 course. The PostgreSQL database is hosted separately (managed database or within the same VM), and the Frontend is a static build that can be served via Nginx or a Cloud VM.
Backup	Implement daily backups for critical operational data (Missions, Prescriptions, Spray_Logs, Invoices) using an automatic pg_dump mechanism; perform weekly backups for the entire database as a long-term archive. Backups are kept for a minimum of 30 days and stored on separate storage from the main server to prevent data loss in case of server failure.
Monitoring	Monitor key operational error groups: • Login errors — number of failed login attempts, unusual account suspensions. • Assignment errors — Drone/Operator assignments failed due to scheduling conflicts or equipment unavailability. • Report/Log errors — Incomplete or failed recording of Spray Log/Operation Log. The system records these errors. System_Audit_LogsThe application can also integrate email/Slack alerts for the administrator when the error rate exceeds a certain threshold.

Support	The administrator is the point of contact for receiving and categorizing support requests into three priority levels: Critical (system downtime, data loss), High (operational errors affecting performance such as incorrect assignment, inability to log), and Normal (requests for information correction, usage questions). Requests are recorded through internal support channels (ticket/email) and forwarded to the appropriate role for processing (technical, business).
Maintenance	Routine maintenance includes: updating the Drone/Battery master data, updating the list of users and roles, adjusting Business Rules (e.g., prohibited chemicals list, maintenance alert thresholds), and patching software bugs based on feedback from the testing phase (item 11). Updates are performed outside of peak operating hours to avoid service disruptions.

13. Design Traceability Matrix

FR ID	UC ID	Use Case / Business Process	UI Screen ID (Matches 21 UI Screens)	Database Table	Test Case ID
FR-01	UC-01	Manage User Accounts	UI-01 User Management	Users	TC-03
FR-02	UC-02	Monitor System Activities and Audit Logs	UI-02 System Audit Log	System_Audit_Logs	TC-04
FR-03	UC-03	User Login and Secure Authentication	UI-03 Login Screen	Users	TC-01, TC-02
FR-04	UC-04	Manage Field Maps and Land Profiles	UI-04 Field & Map Management	Fields	TC-05
FR-05	UC-05	Request Spraying Service	UI-05 Service Request	Service_Requests	TC-06, TC-07
FR-06	UC-06	Automatically Detect Scheduling Conflicts	UI-06 Mission Scheduling	Missions	TC-12
FR-07	UC-06	Display No-Fly Zone (NFZ) Map	UI-06 Mission Scheduling	No_Fly_Zones	TC-10
FR-08	UC-06	Display UAV Resource Availability	UI-07 Monitoring Dashboard	Drones, Batteries	TC-10
FR-09	UC-06	Monitor Mission Progress	UI-07 Monitoring Dashboard	Missions	TC-10
FR-10	UC-07	Create Electronic Prescription	UI-08 Electronic Prescription	Prescriptions	TC-08

FR-11	UC-07	Validate Restricted Chemicals	UI-09 Restricted Chemical Validation	Banned_Chemicals	TC-09
FR-12	UC-09	Manage Drone and Battery Inventory	UI-10 Drone & Battery Management	Drones, Batteries	TC-15
FR-13	UC-10	Manage Drone Maintenance	UI-11 Maintenance Management	Maintenance_Records	TC-16
FR-14	UC-10	Generate Automatic Maintenance Alerts	UI-11 Maintenance Management	Maintenance_Records	TC-16
FR-15	UC-11	Verify Drone Operator License	UI-12 Operator License Verification	Operator_Licenses	TC-13
FR-16	UC-12	Complete Pre-flight Safety Checklist	UI-13 Pre-flight Safety Checklist	Preflight_Checklists	TC-17, TC-18
FR-17	UC-13	Operate Mobile Application Offline	UI-14 Offline Mobile Application	Missions (Local DB Cache)	TC-19
FR-18	UC-14	Record Spray Log	UI-15 Flight & Spray Log	Spray_Logs	TC-19
FR-19	UC-14	Record Operation Log	UI-15 Flight & Spray Log	Operation_Logs	TC-19
FR-20	UC-08	Report Emergency and Update Emergency Status	UI-16 Emergency Alert	Emergency_Reports	TC-14
FR-21	UC-15	Track Mission Status	UI-17 Order Tracking	Missions	TC-20
FR-22	UC-16	Manage Electronic Invoices	UI-18 Electronic Invoice	Invoices	TC-21
FR-23	UC-16	Record Manual Payment Confirmation	UI-19 Payment Confirmation	Invoices	TC-22
FR-24	UC-17	Submit Service Review	UI-20 Service Review	Reviews	TC-23
FR-25	UC-18	Generate Revenue Reports and Export Data	UI-21 Reporting Dashboard	Invoices, Missions	TC-24

14. AI Audit Log

This section documents the use of AI tools to support the team during the development of the System Design Specification (Deliverable 4) — including System Architecture, Technology Stack, Security Design, and QA/Test Strategy for the Agriculture UAV system. Each entry shows the prompt, a summary of the AI output, and the team's Decision & Refinement decisions based on the project's practical constraints (time, resources) and consistency with previous Deliverables.

STT	Week / Phase	DTC	AI Tool	Prompt	AI Output	Decision & Refinement	Verification
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1	W8 / Phase 4 — Design	Pattern Recognition	Claude	Comparison of 3-Layer Modular Monolithic and Microservices architectures for Agriculture UAV systems, project scope ~10 weeks, 1 team of 5 members.	A general comparison based on criteria such as scalability, complexity, and cost—the default conclusion favors Microservices due to their "higher scalability".	Do not follow the AI's default conclusions. We chose 3-Layer Modular Monolithic because it fits within the 10-week development timeframe and the resources of one team — the infrastructure/monitoring costs of Microservices (API Gateway, Service Discovery, Container orchestration) are not commensurate with the scale of an academic demo.	Critical Thinking: AI tends to suggest "trending" architectures regardless of the project's actual time/resource constraints — the team reassesses based on 7 specific criteria (item 7 Architecture Trade-off Analysis) before making a decision.
2	W8 / Phase 4 — Design	Decomposition	ChatGPT	The proposed Technology Stack is suitable for the chosen 3-Layer architecture (Presentation, Application, Data), prioritizing common technologies that are easy for the student group to implement.	AI suggests React for Presentation, Spring Boot for Application Layer, PostgreSQL for Data Layer, and JWT for validation.	Accept the entire proposed stack. Adding the option of Spring Data JPA as an intermediate data access layer (instead of pure JDBC) to align with the Data Flow Overview description already provided in section 6 — AI only suggests PostgreSQL at the database level, not a specific ORM layer.	No in-depth technical verification is needed — the stack is a popular choice, and the instructors have appropriately suggested it during the architectural review session.

3	W9 / Phase 4 — Design	Process /Algorithm	Claude	The system's security design includes authentication and authorization mechanisms based on RBAC with six roles (Administrator, Manager, Farmer, Agronomist, Drone Operator, Technician), and data protection.	The AI suggested JWT + BCrypt for authentication, a 6-role RBAC matrix for access scope, and recommended HTTPS/TLS for overall communication .	Accept all Authentication and Data Protection proposals. One important constraint that AI didn't mention is that permission checks must be performed at the Backend (Spring Security) layer, not just hidden/visible on the Frontend interface — to prevent users from directly calling the API to bypass UI controls, a common security vulnerability that the original prompt didn't require AI to consider.	Compare the RBAC matrix with the relevant FR permission list in Deliverable 2 — ensuring no role is granted permissions exceeding the specified FR.
4	W9 / Phase 4 — Design	Pattern Recognition	Gemini	Develop a Threat & Mitigation Matrix for agricultural UAV systems, identifying common threats to agricultural data management systems and drones, along with their priority levels and mitigation measures.	AI đề xuất 6 threat chung: unauthorized access, SQL injection/XSS, brute-force, data leakage, tampering log, loss of data.	Accept 6 generic threats. Two additional threats specific to the agricultural UAV domain that the AI did not automatically propose are: misassigning UAVs/operators to missions (T-02, posing operational risks, not just data security risks) and violating no-fly zones/prohibited chemicals (T-08, legal risks — industry-specific, not a typical IT threat).	Review all 8 threats against the Business Rules (Deliverable 2) list and the No_Fly_Zones/ Banned_Chemical data entity — verify that the corresponding mitigation matches the specified business rule, not just a general technical solution.

5	W9 / Phase 4 — Design	Decomposition	Copilot	Propose a QA and Test Strategy for the system, based on the Use Cases specified in Deliverable 2 and 3, focusing on Functional Testing and Negative Testing.	AI proposes a three-tiered testing strategy: Unit Test, Integration Test, and Functional Test based on Use Cases, along with a sample test case checklist for UC-05.	Accept the 3-tier test structure. Narrow the scope of Unit Tests/Integration Tests in the SDS document to describe the strategy, not detailed test case implementations — because the scope of Deliverable 4 is System Design Specification, not yet at the stage of writing actual test code.	Compare the proposed scope with the Diagram List and the scope declared in section 1.2 (In-scope/Out-of-scope) — ensuring it does not exceed the scope of the subject.
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